

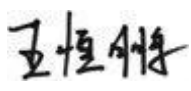
PROJECT TITLE: TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT
AND SIDESTREAM NITROGEN REMOVAL
FACILITIES

CLIENT:



CONSULTANT:



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TUAS WRP

DTSS 2

CONSTRUCTION OF TUAS WATER RECLAMATION
PLANT (TWRP)

BIM Execution Plan

Prepared for



Jacobs

Submission Date: 14/08/2024



CES_SALCON-SDC JV

Prepared by:

Contract Name: Tuas Water Reclamation Plant Contract 4B –
Digestion Pre-Treatment and Side Stream Nitrogen Removal Facilities
Contract Number: DTSS2/33220081
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Acronyms and Abbreviation

AIM	Asset Information Model
ASM	Asset Management System
BIMM	Building Information Management Manager
BIMT	Building Information Management Team
BIR	Board's Information Requirements
BoQ	Bill of Quantity
CAD	Computer-Aided Design
COBie	Construction Operations Building Information Exchange
CDE	Common Data Environment
CSA	Civil, Structural, and Architectural disciplines
EDMS	Electronic Document Management System
EIR	Employer Information Requirements
FM	Facility Management
IFC	Issue for Construction
LoD	Level of Detail
LoI	Level of Information
LOD	Level of Development
MEICA	Process Mechanical, Electrical, Instrumentation, Controls and Automation disciplines
MEP	Non-Process Mechanical, Electrical, and Plumbing disciplines. Also referred to as Building Services)
MIDP	Master Information Delivery Plan
OBD	Open Building Designer
OPM	Open Plant Modeller
OPPID	Open Plant P&ID
O&M	Operations & Maintenance
PCM	Project Controls Manager
PIM	Project Information Model
QS	Quantity Surveyor
QTO	Quantity take-off

TWRP	TUAS Water Reclamation Plant
VCM	Virtual Construction Model
VDC	Virtual Design and Construction
VDM	Virtual Design Model
.ifc	Industry Foundation Classes

Glossary

As-Built	The record drawings and documentation define deviations to the designed information occurring during construction at the end of the project.
As-Constructed Model	Also referred to as Construction Progress Model information during construction. The “As-Constructed” model and its appended documentation are continually updated as construction progresses.
Asset Information Model	Maintained information model used to manage, maintain, and operate the asset after construction.
BIM Guide	A guide to the use of BIM published by the Building and Construction Authority (The Singapore BIM Guide – Version 2).
BIM Model	A digital representation of the physical and functional characteristics of the project. The model comprises 3-dimensional graphical information, non-graphical information, and documentation.
Building Information Modelling	A process of designing, constructing, or operating a building or infrastructure asset using electronic object-oriented information.
Clash Rendition	Representation of the native format model file, used specifically for spatial planning and coordination processes. This is to achieve clash avoidance through clash analysis.
Classification	The action or process of categorizing, grouping, or sorting elements according to shared qualities or characteristics.
Common Data Environment	Single source of information for any given project used to collect, manage, and disseminate all relevant project artifacts (including 3D models, drawings, documents, etc.) for multi-disciplinary teams in a managed process.
Design Authoring	A process in which BIM software is used to develop a Building Information Model based on criteria that are important to the translation of the building’s design.

Electronic Document Management System	System for storing, retrieving, sharing, and otherwise managing electronic documents in line with BS1192.
Handover Model	As defined in the BCA VDC Guide, the version of the Model changes ownership and authorship. (i.e., The Released Model of design intent that is handed over to the awarded Contractor for their visual reference and further development).
Industry Foundation Classes (File extension)	A data model intended to describe building and construction industry data in a platform-neutral and open file format.
Level of Development	The “Level of Development” is the degree to which the elements graphical and non-graphical (attributes) have been thought through a particular stage.
Level Of Detail	Description of graphical content of models at each of the stages (PAS 1192-2:2013)
Level of Information	Description of non-graphical content of models at each of the stages (PAS1192-2:2013)
Master Information Delivery Plan	The primary plan for when project information is to be prepared, by whom, and using what protocols and SMP (standard methods & procedures), incorporating all relevant Task Information Delivery Plans (TIDP).
Major Changes	Defined as significant changes that alter the original intent and normally involve design changes like addition, subtraction, and modification.
Minor Changes	Defined as acceptable tolerance deviations, or movement of elements through horizontal and vertical offsets that do not require major remodelling works.
Project Information Model	As defined in BS1192, the information model is developed during the design and construction phase of a project. PIM comprises a series of individual models, federated models, and non-graphical information developed progressively during the project.
Released Model	As defined by BCA BIM Particular Conditions, this means a Model identified as such and released by the Model Author from time to time as agreed in the BIM Execution Plan.
Virtual Construction Model	Subsequent version of the project information model developed from the VDM by the contractor and their supply chain.

Virtual Design Model	This is the version of the Project Information Model (PIM) developed by the Consultant (for CSA) and Contractor (for MEP and MEICA elements).
4D	Construction Sequencing.
5D	Quantities and Cost Tracking.
6D	Asset Management.

1 Purpose and Objectives

1.1 Purpose

1.1.1 Project Information

This Project Information is shown in Table 1-1 below.

Table 1-1 Project Information

Contract Name	Tuas Water Reclamation Plant Contract 4B – Digestion Pre-Treatment and Sidestream Nitrogen Removal Facilities
Contract Number	DTSS2/3322008
Project Address	Tuas View Basin, Tuas District, Singapore
Scope of works	The major work scope to be provided under Contract 4B includes (i) Digestion Pre-treatment Facility; (ii) Sidestream Nitrogen Removal Facility; (iii) Sitewide pipe rack and interconnecting pipework between Biosolids area and ILM; and (iv) Interfaces between the facilities and Sitewide, ILM 1 facility, Biosolids facility and IWMF.

1.1.2 Purpose

This BEP will provide guidelines that ensure CES project team will receive, produce and release data in a consistent format for TWRP C4B project. This will maintain an efficient exchange of data between disciplines and contracts and the compatibility of each discipline and contract's model(s).

The primary purpose of this BEP is to standardize the BIM work progress inside CES project team and improve the coordination among CES project team with other functional groups within TWRP C4B, to design a suitable process to apply BIM in C4B project execution. , as well as to develop C4B BIM work in a way that will facilitate the further use of this electronic information beyond the initial contract.

1.2 BIM Objectives

The purpose of this table is to document the main objectives of using BIM to deliver the Works and the high-level methodology for achieving each objective.

Table 1-2 Project BIM Objectives

Goal	Objective
Aim to facilitate better collaboration among various stakeholders involved in a construction project, including architects, engineers, contractors, and owners	Leverage model visualizations and intelligent data to support decision-making processes and present information in a fashion that is easily understood.

Goal	Objective
Provides a common platform for sharing information, reducing misunderstandings and conflicts between different parties.	Enhance collaboration and communication amongst all stakeholders to improve the quality of models and ensure timely project implementation.
Enhance worker productivity due to better-coordinated information	Ensure that all disciplines' models are coordinated to prevent clashes and conflicts between different building systems.
Mitigate project risk to Board in the form of disconnected data and inaccurate models	Drive the process of information and model management to eliminate cost overruns at the project site where coordination is exponentially more expensive than the virtual environment.
Through accurate and detailed virtual modelling, BIM helps identify potential issues before they become costly problems on the construction site. Enables better project planning, scheduling, and coordination, leading to improved project management and reduced delays	Integrate intelligent model data with cost and scheduling activities. This will increase the ability to efficiently review costs for multiple options and understand potential schedule conflicts for complex construction activities.
Oversight of actual construction progress vs. virtually planned work	Simulate the complex construction sequence to identify and communicate any potential risks virtually, first.
Control and streamline information for seamless turnover and best-in-class Asset Management environment	Create a controlled data environment throughout the life cycle of the project to facilitate a digital exchange of operations and maintenance information at project completion. Mitigate the painful process of manual information input at the close of the program. Reduce the risk of maintenance and operations items that may require early attention while information is being inputted.
Use new and innovative technologies where appropriate	Select and implement the appropriate BIM software tools for the project's needs. This includes training team members to effectively use the software.
Allows for better visualization and analysis of design concepts, resulting in higher-quality design solutions	3D models can reflect the design intuitively, which can help to check and improve the design.

Goal	Objective
Provide a container for all the information produced during the design-construction stage	3D model can serve as a container to store and exchange the project datadrop.

The following table defines the BIM uses for the model and the methodology to achieve those uses.

Table 1-3 BIM Uses

No.	BIM Use	Methodology
1	Preliminary Design	
1.1	Regulatory Submission	Develop & maintain the Model according to the Preliminary Design
1.2	Identify element conflicts and interferences	Clash Detection and Resolution Report
2	Detail Design	
2.1	Regulatory Submission	Develop & maintain the Model according to the Detail Design
2.2	Identify element conflicts and interferences	Clash Detection and Resolution Report
3	Construction	
3.1	Identify element conflicts and interferences	Clash Detection and Resolution Report
3.2	Shop Drawings	Generate Shop Drawings based on the Construction Models
3.3	Construction Sequencing (4D)	Map 3D objects to the date/time as shown in the P6 Programme and simulate through Synchro
3.4	Cost estimation (5D)	Attach the model with quantification schedules
3.5	Asset Management data turnover (6D)	Prior to completion of construction, embed Construction Operations Building Information Exchange (COBie) and PUB's asset management data on all model elements that represent "Managed Assets"
4	As-Built	

4.1	Final As-Built models fit for space management, building maintenance	Incorporate operation and maintenance information of major systems and equipment in the BIM model elements
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To further elaborate on how CES BIM Team will utilize BIM in the project, here are some additional points:

- a. **Project Coordination:** CES BIM Team will use the BIM model as a central hub for project coordination. CES BIM Team will integrate design information, construction schedules, and cost data within the model to facilitate collaboration among various project stakeholders. This enables effective communication, clash detection, and resolution of conflicts during the construction process.
- b. **Detailed Construction Sequencing:** CES BIM Team will visualize and simulate the construction sequencing, the 4D simulation works can be done not only based on a monthly schedule but also for some dangerous construction items to avoid losses. This enables the Project Team to optimize construction activities, identify potential conflicts or logistical issues, and streamline the construction process for improved efficiency.
- c. **Material Take-off and Quantity Estimation:** The BIM model provides the BIM Team with the ability to perform accurate material take-offs and quantity estimations. By extracting information directly from the model, CES BIM Team can generate precise quantity reports, which aids in procurement planning, cost estimation, and resource management. This helps to reduce errors and improve cost control.
- d. **Design and Constructability Reviews:** CES BIM Team will utilize the BIM model to conduct design and constructability reviews. By analysing the model, they can identify potential design issues, clashes, or constructability challenges early in the process. This allows for proactive problem-solving, reducing rework and avoiding costly delays during construction.

By leveraging BIM in these ways, CES BIM Team will enhance project coordination, streamline construction processes, improve cost control, reduce errors, and support effective facility management. BIM will serve as a powerful tool throughout the project lifecycle, enabling the C4B project to achieve better project outcomes in terms of quality, efficiency, and collaboration.

1.3 Information Required by EIR

The following table defines the minimum contents of the Employers Information Requirements (EIR) and the response subsections of the BEP to the parts of the EIR.

Table 1-4 Minimum Contents Required by EIR

EIR Section No.	Summary of Requirements	In this BEP
P2C.9(d) (i)	Project Information	Section 1.1
P2C.9(d) (ii)	BIM Objectives and goals	Section 1.2
P2C.9(d) (iii)	BIM Uses	Section 1.2

EIR Section No.	Summary of Requirements	In this BEP
P2C.9(d) (iv)	Governance, roles, and responsibilities	Section 2
P2C.9(d) (v)	BIM protocols	Section 6.1.1
P2C.9(d) (vi)	Model development approach	Section 3
P2C.9(d) (vii)	Level of Development	Section 4.2
P2C.9(d) (viii)	Procedures, methods, collaboration, and training	Section 3 Section 4.1 Section 5 Section 8.3
P2C.9(d) (ix)	Supply chain	Section 7.3
P2C.9(d) (x)	Master information delivery plans (MIDP)	Section 9 Appendix – A.9
P2C.9(d) (xi)	Interface management	Section 7
P2C.9(d) (xii)	BIM compliance plan	Section 6.1

2 BIM Execution and Management

2.1 BIM Organization Chart

BIM team includes a BIM manager, a BIM Authoring Lead, a Database Lead, and a BIM Modelling Team.

The detail of the Project and BIM organization is shown in the chart below.

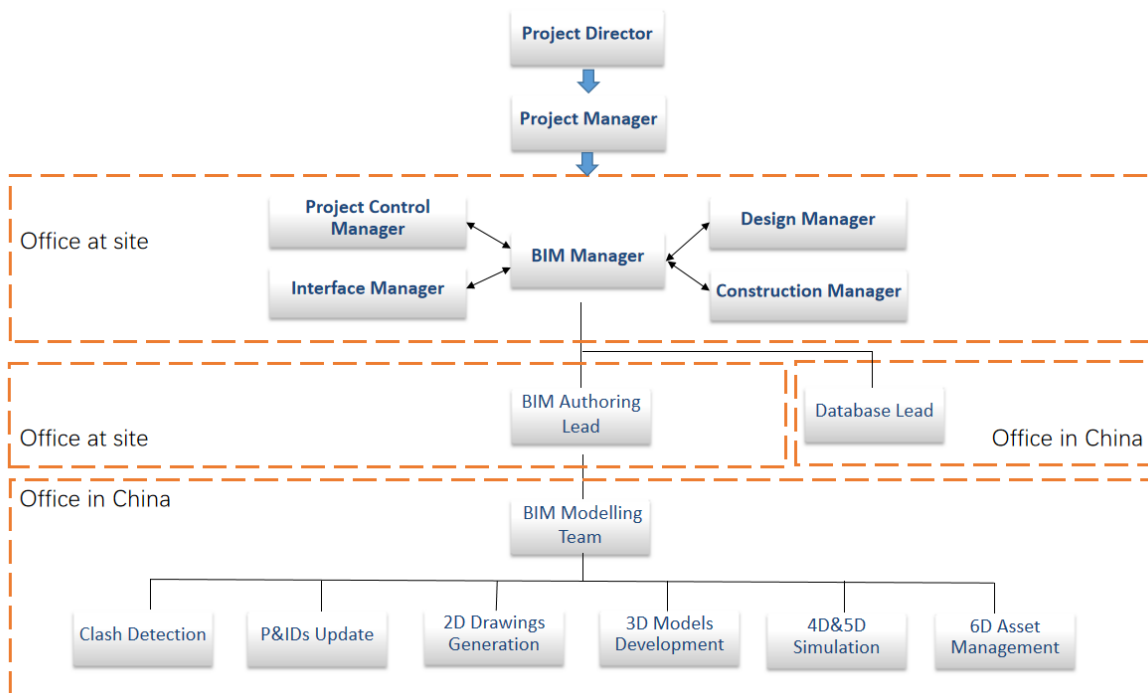


Figure 2-1 C4B Organization Chart

2.2 Project BIM Contact

The table below shows the details of project member contact relating to BIM work.

Table 2-1 Details of Project Member Contact

Role	Contact Name	E-Mail Address	Phone
BIM Manager	Wang Heng Peng	Wang_hp2@hdec.com	+65-85438328
Database Lead	Pan Yiru	pan_yr@hdec.com	+86-13646713976
BIM Authoring Lead	Muthukumar Jayabalan	muthu_er@yahoo.co.in	+65-97204013

2.3 Roles and Responsibility

- 1) BIM Manager
 - a) Under the leadership of the project manager, and interface with the Design team, Construction Manager, Interface Manager
 - b) Lead the creation of BIM deliverables such as BIM Execution Plans, BIM Quality Control Plans, BIM Design Review Procedures, Construction Sequencing (4D), and Cost Estimating (5D).
 - c) Manage BIM standards, resources, and application configuration.
 - d) Lead BIM teams (both on-site and off-site). Manage the BIM teams to fulfil the project deliverables on schedule.
 - e) Attend all BIM meetings when required.
 - f) Responsible for 6D model production.
 - g) Responsible for preparing the BEP and modifying it according to the comments until the BEP is approved.
 - h) Serve as a single point of contact for the S.O.
- 2) Database Lead
 - a) Responsible for setup and configuration of the P&ID database.
 - b) Provide support during the Contractor design and construction of MEICA elements.
- 3) BIM Authoring Lead
 - a) Responsible for the ongoing updating of the model to construction and As-Built status.
 - b) Lead and manage BIM modelling team.
 - c) Coordinate with other departments of CES for BIM work.
 - d) Assists in the production of BIM deliverables.
 - e) Check model quality and integrity before the generation of deliverables.
- 4) BIM Modellers
 - a) Generate 2D drawings from the model.
 - b) Maintain and develop the BIM model.
 - c) Clash check and assist BIM Lead to solve the clashes.
 - d) Update the PID drawings.
 - e) Prepare the 4D & 5D simulation and 6D asset management.
 - f) Annotate model views by adding dimensions, tags, legends, schedules, etc.

3 BIM Approach and Processes

3.1 Model Production and Authoring

This section specifies the workflows for the main BIM works carried out during Preliminary Design Stage, Detail Design Stage, Construction Stage and As-Built Stage, in order to achieve qualified deliverables. Related works below include P&ID development, model development, model review procedure, construction coordination, Health and safety works with BIM, 4D works, 5D works and As-Built work.

3.1.1 P&ID Development Workflow

In the preliminary design and detail design stage, prior to the creation or further updates of the model, it is necessary to first design P&ID drawings. The design team should coordinate and review the P&ID work internally, and then provide comments on the P&ID based on the .pdf P&ID drawings to the BIM team, like marking the pipes routine, and valve's location adjustment on the drawing according to the design. CES BIM team then update the P&ID drawings according to the comments. Meanwhile, the design team should provide datadrop information to the BIM team in the form of .xlsx file. The BIM team then inputs the received datadrop information into the database and update the datadrop into the P&ID drawings. Finally, The P&ID drawing and all list reports should be reviewed by Engineering Manager, Quality Manager, and BIM manager, and the review work should follow the checklist 131242-WST00-FM-IM-0006, and the P&ID results are submitted for review to S.O.. The final deliverables should include drawings in .dgn, .i.dgn, and .pdf formats, a datasheet report, and checklist in .xlsx format. The approved P&ID drawings will serve as the basis for further updating the model.

During P&ID drawing creation, if new items are needed which are not in the latest OPPID dataset, the Contractor needs to discuss this internally between BIM team and design team to confirm these new items are needed. These should be raised to S.O. to clarify in BIM coordination meeting. Then official dataset change request should be submitted to S.O. for OPPID dataset update by contractor.

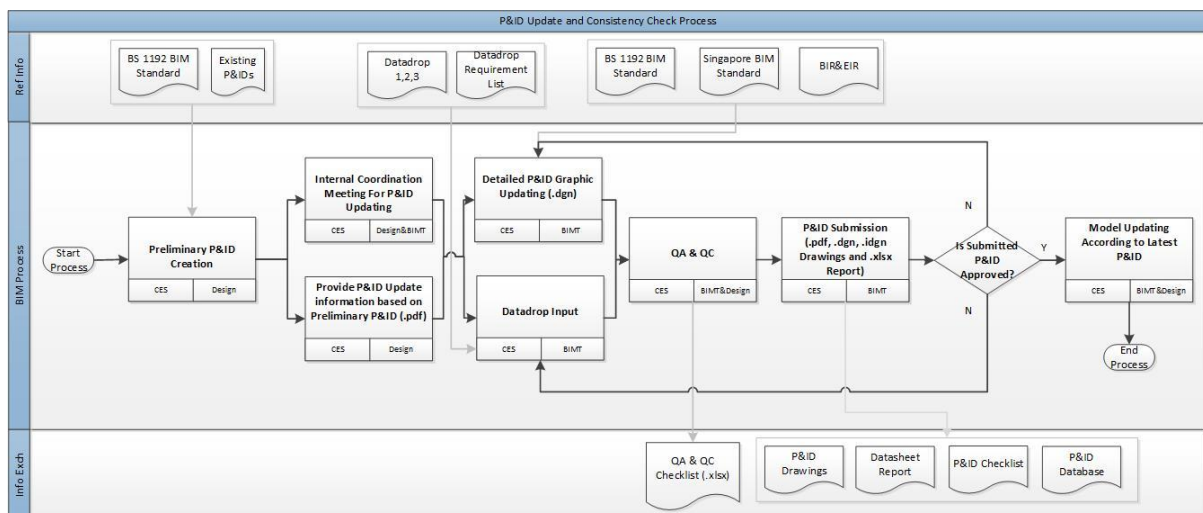


Figure 3-1 P&ID Development Workflow

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.1.2 Preliminary Design Model Production and Authoring

In the preliminary design stage, the design team will provide preliminary design proposals for the CSA, MEICA, and BS disciplines, like preliminary design drawings and preliminary P&ID drawings. The BIM team will create the preliminary design model and update to LOD200 based on the project's BIM specification requirements and the design inputs confirmed by the design team and manage the content division and naming of each discipline's sub-models.

For CSA discipline, main CSA elements should be modelled with true size as design input and preliminary consideration should be given to cast-in items. For MEICA discipline, special attention is given to checking the consistency between the model and P&ID drawings, ensuring that the P&ID design drawings are first determined before the model is built according to the standard process. For BS discipline, the main route should be updated based on the design input and consider the preliminary arrangement of MEICA items. After all discipline design models are created and updated, and no more preliminary design changes, BIM team should perform a clash check on all discipline models. For any clash found, BIM team need to discuss them with the design team and obtain necessary approvals through the S.O. process. After resolving all model clashes, BIM team will conduct milestone 1 model review (As depicted in Section 3.1.6) with S.O., update the model with S.O. comments and submit the preliminary design model to the S.O. for approval.

In the preliminary design stage, elements should be modelled with preliminary tagging information. All 3D geometry will have a baseline level of intelligence. A specific and descriptive family name and type name at a minimum will be used to describe the object. This will allow for a more effective use of the BIM data downstream.

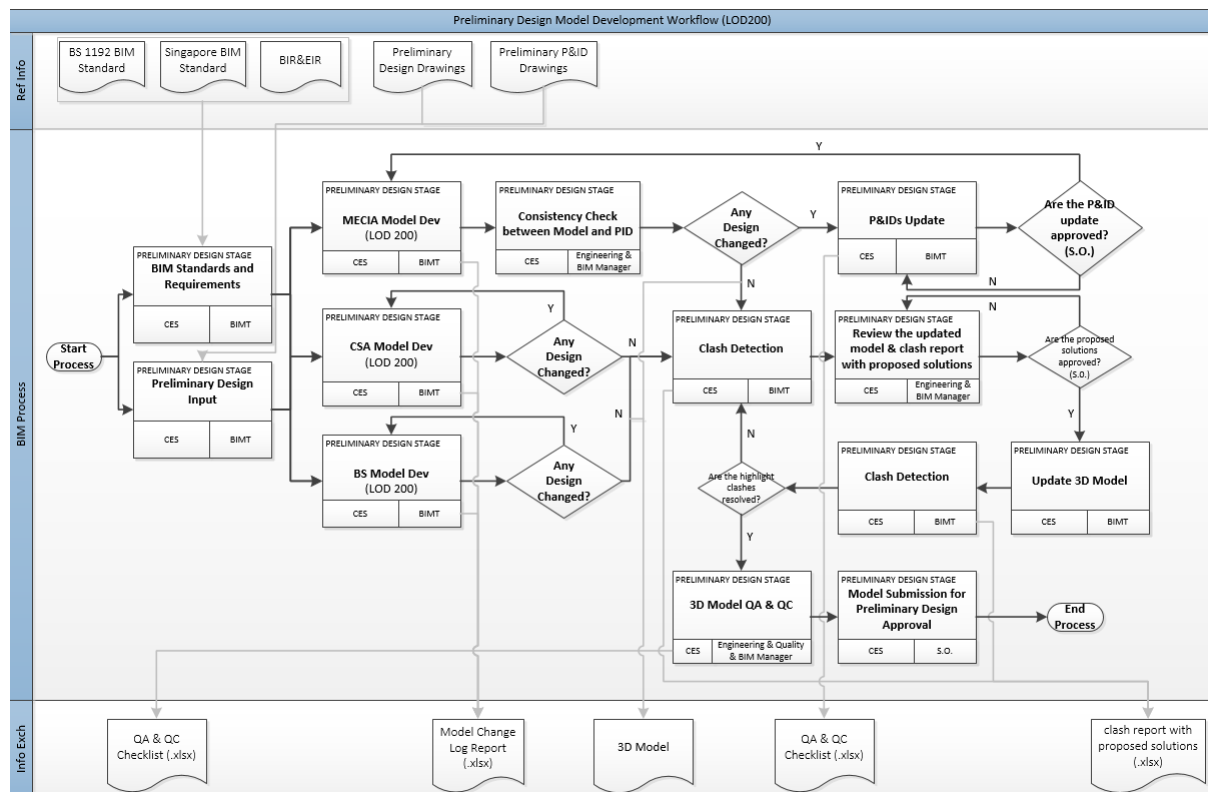


Figure 3-2 Preliminary Design Model Development Workflow (LOD200)

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.1.3 Detail Design Model Production and Authoring

In the detail design stage, BIM team will identify information requirements for the construction model from standards, EIR, and BIR. All discipline's design will be optimized and detailed modelling work will be carried out based on the preliminary design model. Detail design inputs, such as design optimized drawings, and typical drawings, will be provided based on the review on the preliminary design model.

For CSA disciplines, elements should be modelled with confirmed and accurate sizes and locations. For MEICA disciplines, P&ID design should be updated and get approval first. Then elements should be updated as latest approved P&ID design. For BS disciplines, models should be updated with details of branch items. Any model changes will be recorded in the Model Change Log Report (.xlsx). After all models are completed with detail design, BIM team will conduct clash detection analysis, record in clash reports and review the clashes with engineer team and S.O., until all clashes are resolved. Then BIM team will conduct milestone 2 walkthrough (As depicted in Section 3.1.6) with S.O. After models are updated based on S.O comments. BIM team will prepare authority drawings with models for S.O. review. Then the authority drawings will be submitted for approval. Once the authority drawings are approved, CES team will further prepare shop drawings .

In the detail design stage, datadrop 1 will be input to assist engineer team to do model review.

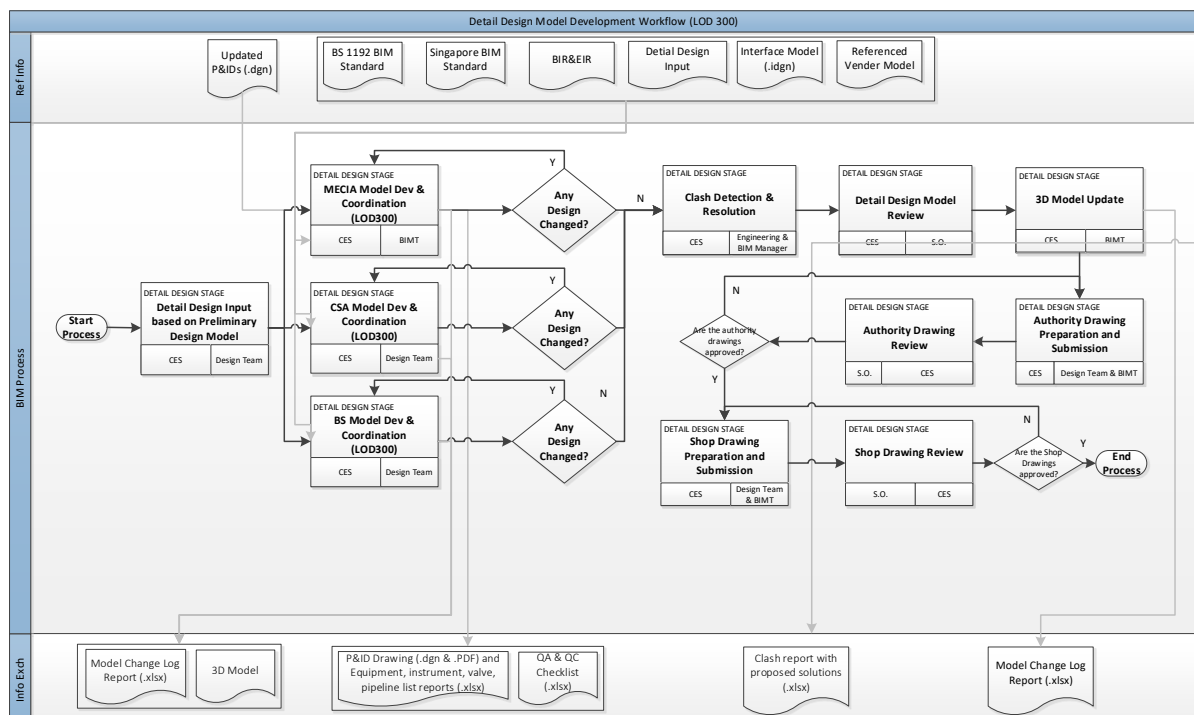


Figure 3-3 Detail Design Model Development Workflow (LOD300)

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

Table 3-1 Authority Submissions

N/ A	AUTHORITY	
1	URA	Urban Redevelopment Authority
2	BCA	Building and Construction Authority
3	SCDF	Singapore Civil Defence Force
4	FSSD	The Fire Safety and Shelter Department
5	LTA	Land Transport Authority
6	PUB	Public Utilities Board

3.1.4 Construction Model Development

In the construction stage, all disciplines will continue to further develop the models with construction details based on the detailed design model until the model accuracy reaches LOD400, to ensure the model details are enough for shop drawings improvement, quantities extraction, material fabrication, etc.

BIM team will extract 2D drawing from 3D model to prepare Authority Drawing, and base on the approval of Authority drawing, BIM team will continue to develop shop drawing for construction. If it is necessary to resubmit the Authority drawing due to design change, BIM team will continue to develop shop drawing after the resubmission of Authority drawing get approval.

BIM team will keep updating 3D model according to the actual completed works data on site, like laser scanning data, survey data, measurement data from construction team, etc. Any changes especially those influence the design will be reflected in the 3D model and reviewed by engineer and S.O. again. After all changes are processed and all clashes are resolved. Shop drawings will be updated for construction.

In the construction stage, 6D Datadrop 2 will be provided in the model all the applicable data that is available during the installation and construction. Also, 4D and 5D codes will be input into the model and utilized to assist project schedule and cost management (As specified in Section 3.4 and 3.5). Some TWRP Information of model should be also input and updated in time, including shop drawing status, works installed status, works inspected status, etc. The information will be used for construction coordination (As specified in Section 3.2). After all are finished and approved, then this process can be ended.

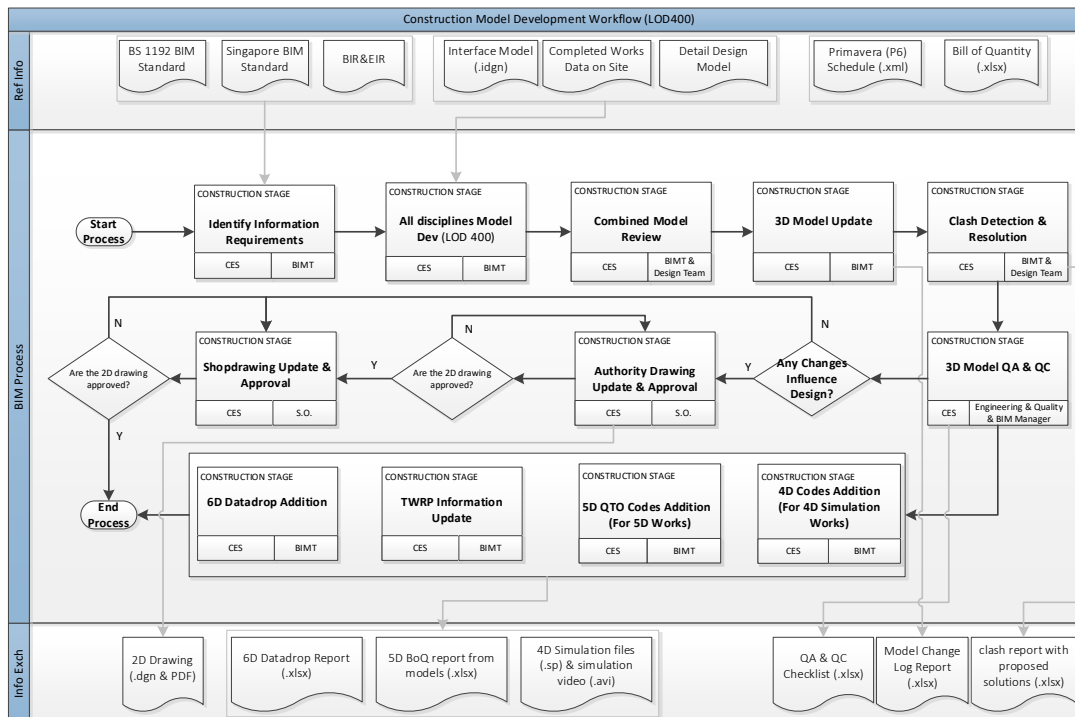


Figure 3-4 Construction Model Development Workflow (LOD400)

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.1.5 As-Built Model Development

In the As-Built stage, the As-Built model should be updated based on construction model and monthly photo survey.

For graphic information, as-built model should be done according to the finally completed works data on site. The updated model should be compared with laser scanning data. After that, clash detection should be carried out, and output the clash report with “zero clash” (.xlsx).

For non-graphic information, datadrop should be input into model in As-Built stage. According to As-Built data and requirements in standards and EIR, we shall embed all Construction Operations Building Information Exchange (COBie) data, including datadrop 1, 2 and 3 on all model elements that represent “Managed Assets”. Finally, all datadrop should be reviewed by project team.

After the graphic and non-graphic information are all completed. The As-Built model should be reviewed by Engineering Manager, Quality Manager, and BIM manager, and output the QA & QC checklist. Finally, we shall publish the As-Built BIM Model with COBie/AMS data in “.ifc2x3”; and provide COBie (.xlsx) workbook to help AMS to realize asset management, output the Asset Management System (AMS) – as per the spreadsheet (“.xlsx”), updated database of P&ID data from OPPID.

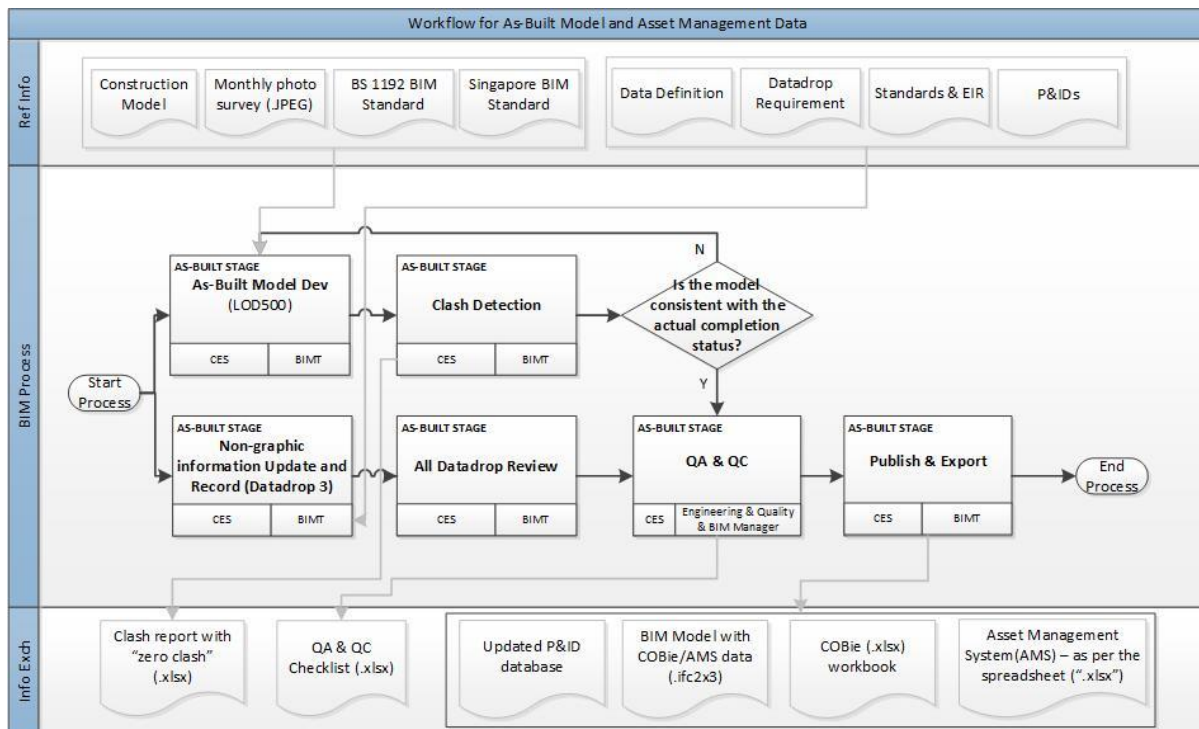


Figure 3-5 As-Built Model Development Workflow (LOD500)

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.1.6 Model Review Procedure

1) Overview

The purpose of this section is to establish a standardized approach for reviewing a 3D Model during the preliminary design and detail design stage of the TWRP C4B project. This procedure aligns with the requirement and recommendation to conduct reviews on components included in the model and layout. It outlines the workflow to be followed during the Detail Design phase model reviews, specifying the scope, timing, responsibilities, and procedures to be adhered to before, during, and after the review process. Additionally, this procedure defines the methods for documenting comments and incorporating agreed-upon resolutions to finalize the Detail Design model. The objectives of the model review are as follows:

- To ensure that the Plant Design adheres to the project requirements.
- To verify the consistency of the model with supporting documents such as P&ID, Equipment Layout Plot Plan, Hazardous Area Classification, specifications, standards, HSE regulations, and local regulations (if applicable).
- To reach a consensus among all involved parties.
- To approve the current design status and provide a path forward.

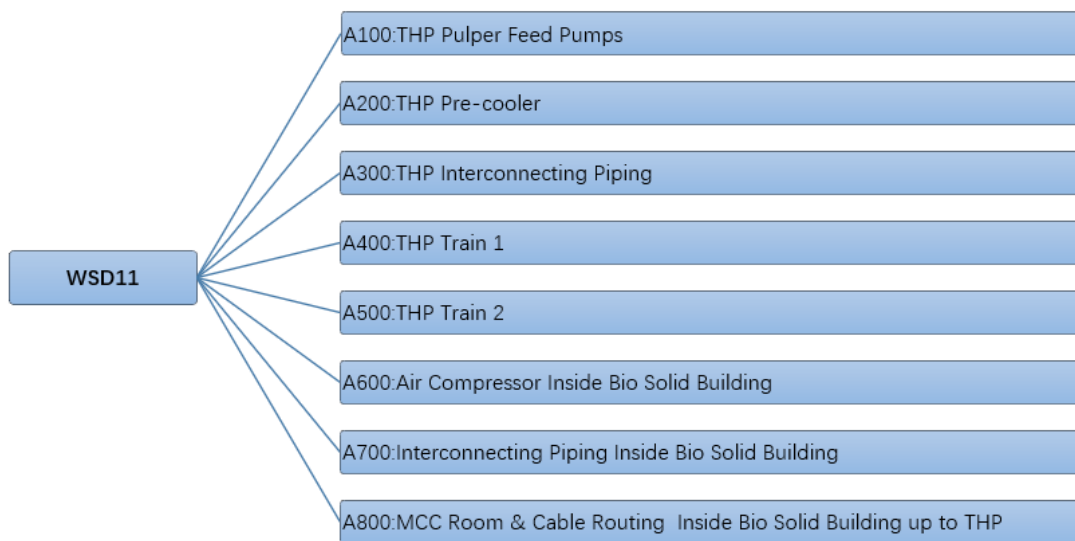
The model reviews will cover the Milestone 1.1, Milestone 1.2, Milestone 2 and Milestone 3 stages of the detail design phase. The review stages are linked to the project milestone and scheduled to maximize the availability of inputs, and engineering information from disciplines (Process, Mechanical, Control systems, etc.).

The Detail Design layout models will be presented through 3D software. This document includes the general procedure to conduct review sessions which shall be structured such that all design and engineering criteria have been met with respect to Company requirements, safety, process, operability, maintainability, and constructability in association with project specification and international codes/standards. A separate model review procedure has also been prepared specific to the building scope due to the magnitude of model review criteria required for the review.

2) Work Breakdown Structure

The project is generally divided into 3 facilities, WSD11 -Digestion Pretreatment, WST02 - Sidestream N Removal, and WSW41 -Sitewide Pipe Rack. Each facility is subdivided into different zones, which will be reviewed as part of milestone process.

a) WSD11 -Digestion Pretreatment



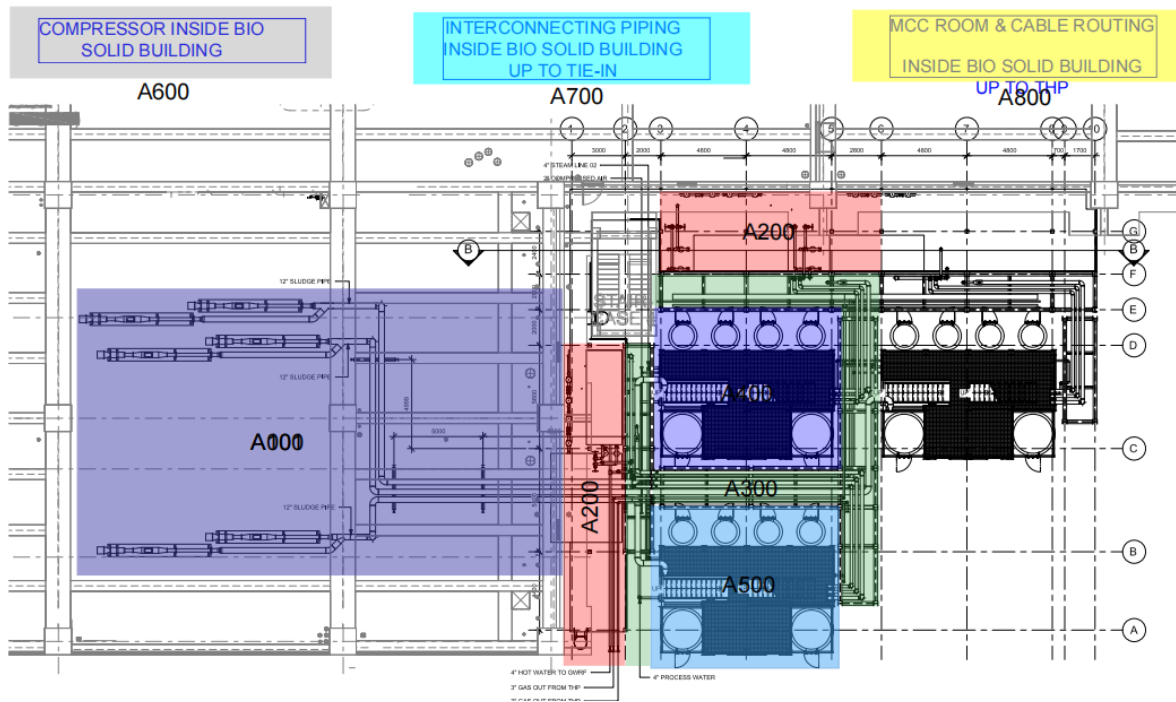
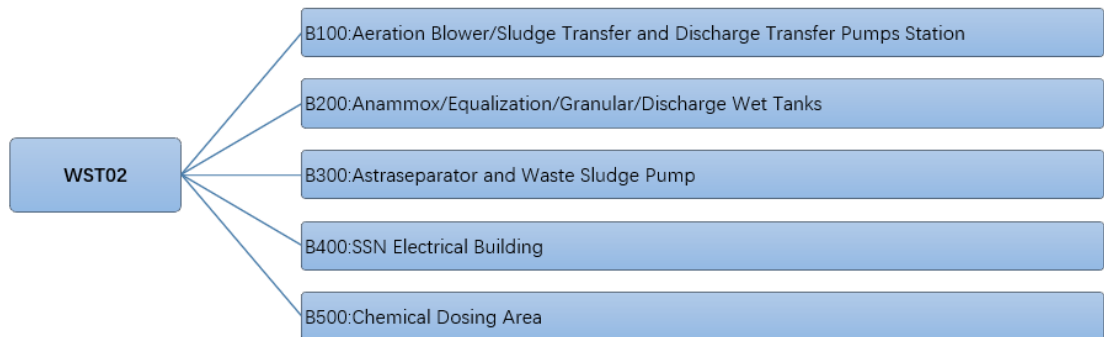


Figure 3-6 Area Demarcation of WSD11

b) WST02 -Sidestream N Removal



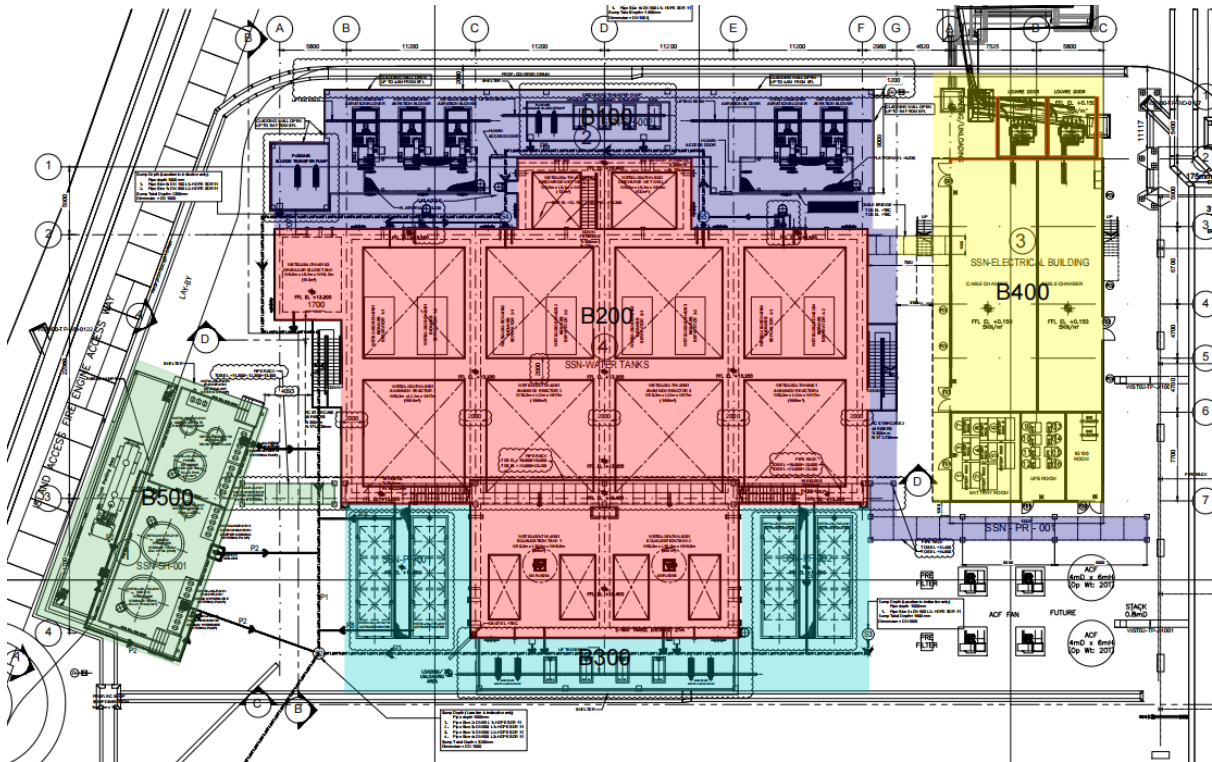
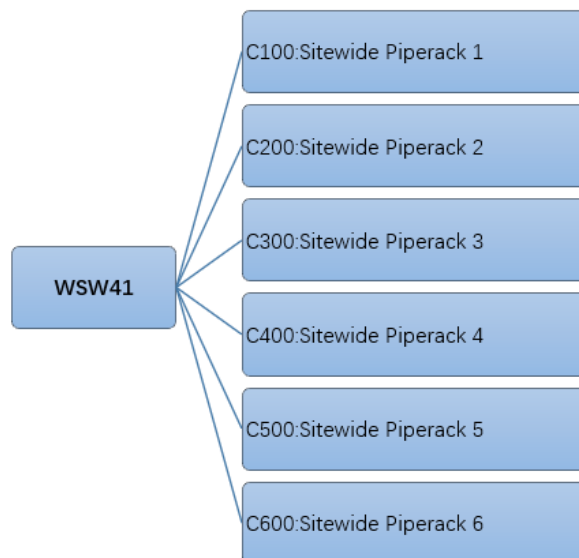


Figure 3-7 Area Demarcation of WST02

c) WSW41 -Sitewide Pipe Rack



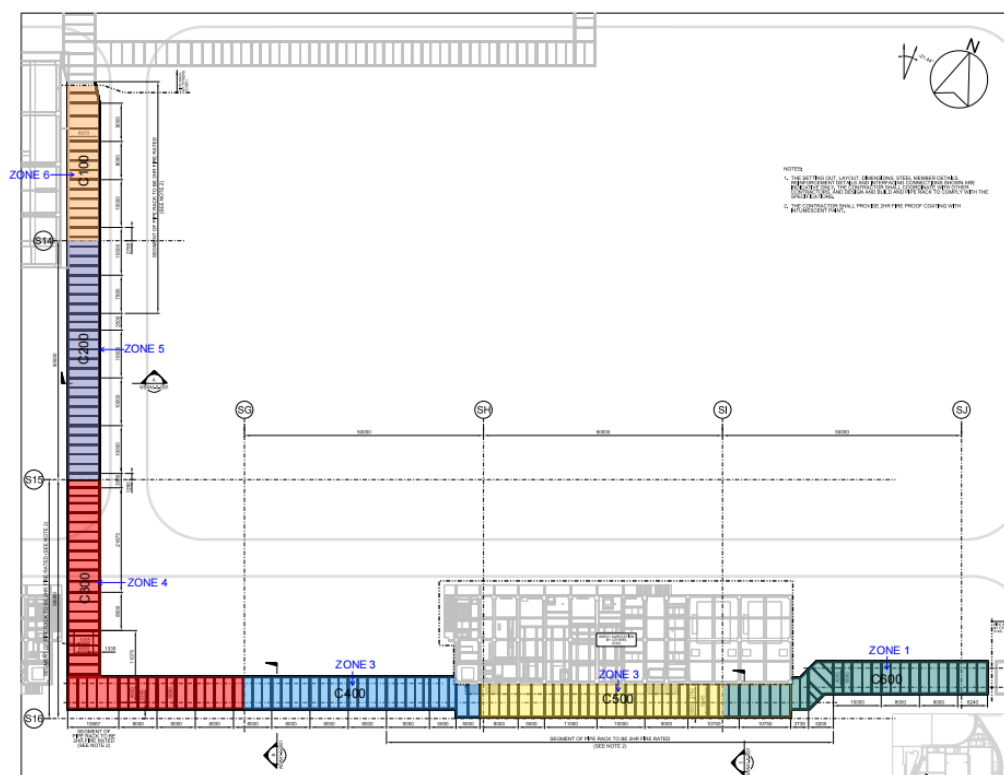
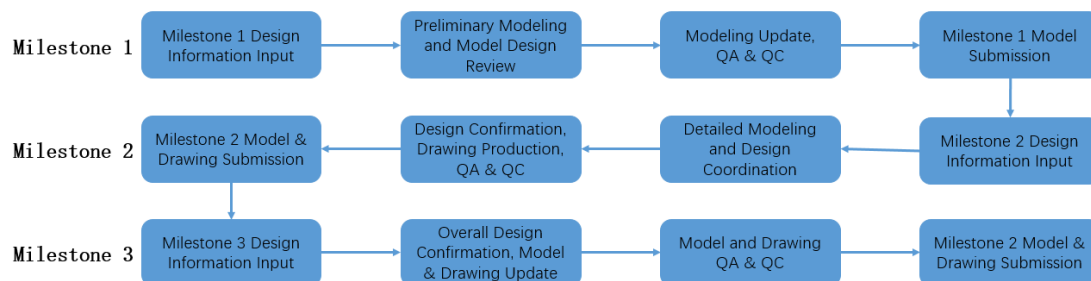


Figure 3-8 Area Demarcation of WSW41

The general workflow of 3D model review is shown below:



*To facilitate layout approval, Milestone 1 is divided into Milestone 1.1 and Milestone 1.2. The input and output of each stage are listed as below.

Stage		Input	Output
Milestone 1 (30%)	Milestone 1.1	Layout Drawing, P&ID Drawing, Preliminary vendor Information,	Milestone 1.1 Model concrete Body Plan
	Milestone 1.2	Layout Drawing(changes), P&ID Drawing(changes), Preliminary vendor Information, Cast-in Location and Size, Datadrop 1, 4D&5D	Milestone 1.2 Model with 4D&5D, Cast-in Items Drawing, Concrete Body Plan
Milestone 2 (60%)		Milestone 1 Model Review Comments, Sketch, confirmed vendor Information, Major Items Confirmation, Support Consideration, ALMOP Consideration, Datadrop 2, 4D&5D	Milestone 2 Model with 4D&5D, 2D Drawing extracted from model

Milestone 3 (90%)	Milestone 2 Model Review Comments, 90% Confirmed Design, 90% Confirmed Vendor Information, Typical Drawing, Datadrop 3, 4D&5D	Detail Design Model with 4D&5D, 2D Drawing extracted from model +Details
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As the review progresses, Model review will be based on the BIM Review checklist (Appendix C.1, 131241-WSW41-CH-IM-0001), all piping, equipment, and instrumentation that has been checked against the model and accepted is to be “yellowed off” on the latest issue of P&ID’s to confirm that it has been reviewed. The review will be on a line-by-line basis, the “to and from” including instrumentation attachment and supports.

Systems shall be reviewed sequentially and methodically with strict attention to detail.

Checking of equipment areas should be considered as discrete blocks. Conclude with checks on safety/ utility and process systems. Any critical systems, highlighted by piping/process engineers, should be subjected to special scrutiny (e.g., high-temperature lines, no pocket lines, etc.).

The model review matrix per section 6.4 has identified the items that will be reviewed during each milestone activity (Milestone 1.1, Milestone 1.2, Milestone 2 and Milestone 3). It shall be followed and completed during the review. A model review slide shall be provided before the date of the actual model reviews for the participants to familiarize of agenda and scope of the review.

Each discipline is responsible for reviewing an area before a general staged review by all disciplines and ensuring they reflect project requirements and that interfaces with other disciplines are correct.

The responsibilities of personnel hosting the model reviews are identified in Section 3.

All associated items in that area will be reviewed and all the comments shall be part of the area.

3) Content

a. Milestone 1-30%

All discipline leads shall review the items listed in the Model Review Subject Matrix. All disciplines shall then resolve all model comments assigned to them in the model review report.

The focus of a 30% model review is on Safety, Constructability, Operability and Maintainability. Human Factors Engineering results should be used as input to the model review. The following is a list of the major expectations upon completion of MS1-30%:

- Agreement on locations of all equipment, buildings, pipe rack and major structures
- Line sizes 300mm and above piping runs, including underground pipe
- Agreement and configuration of critical piping tie-ins for plant shutdown
- Preliminary space allocation for EICA cable trays (Main, preliminary size and route, width 300mm and above)
- Preliminary panel size of 600mm and above and location
- Access ways/Lifting path/Maintenance access/Operational access/Space consumers consideration.
- Escape Routes
- Ergonomic Requirements
- Staging Area
- Agree on major dimensions for Substations
- Cast-in/Concrete slab/Wall openings

The 30% model review shall be concluded with a statement describing which aspects are frozen at this stage.

b. Milestone 2-60%

All discipline leads shall review and validate all model comments assigned in the MS1-30% and confirm that these comments have been resolved and implemented, and the remaining open items on model review matrix are covered.

The focus of a 60% model review is on engineering and technology. The following is a list of the major expectations upon completion of MS2-60%:

- Major items as in approved P&IDs shown in model (include eye wash, safety shower location, except drainpipe, vents, etc.), freeze the location.
- Agreement on process piping configurations (80mm and above)
- Completed stress analysis of critical piping
- Major equipment (pumps, compressors, etc.) sizing completed and shown in model, as per selected equipment dimensions
- Electrical cable trays with 300mm and above (confirmed route and size, no clash)
- Electrical Switchboards (HVSG, LVSG, MCC, and MDB) (Confirmed size and location)
- Equipment + LCS location (Confirmed size and details)
- In-line instrument's locations, accessibility, safe operability, maintainability (control valves, straight run requirement of flowmeter)
- Offline instrument's locations (transmitters, Fire and gas devices and JB's) safe operation, maintainability, and accessibility.
- The ergonomic requirements, escape routes, and the location of the Fire Protection Equipment such as deluge valves, monitors, hydrants, and hose reels will be reviewed.
- Ladders, stairs, railings, and platforms
- Supports for major items (preliminary location and type)
- General piping layout drawing

The 60% model review shall be concluded with a statement describing which aspects are frozen at this stage.

c. Milestone 3-90%

All discipline leads shall review and validate all model comments assigned in the MS1 60% and confirm that these comments have been resolved and implemented, and that the remaining open items on model review matrix are covered. Equipment and in-line instruments modelled to final certified vendor data.

This review also provides an opportunity for a final "walk-through" to validate that the model meets all Company requirements. The following is a list of the major expectations upon completion of MS3-90%:

- All equipment (pumps, compressors, fans, cranes, panels, instruments, cable trays &, etc.) design approved and shown in the model, including minor items such as vents, drains, Eyewash, Safety shower, wash hose station, calibration column, drip trap, etc.
- Complete review of remaining utility piping configuration
- Review of electrical and instrument tray routings
- Review of miscellaneous discipline supports
- Review of any late or special items
- Issue AFC piping isometrics and piping general arrangement drawings
- All clash cases are identified and cleared by S.O with "Accepted to proceed construction".

4) Matrix

This review matrix below identifies the content of the model that would be the basis for the model review and comment process in Detail Design.

SUBJECT INPUT	Discipline	MILESTONE 1 REVIEW (30%)	MILESTONE 2 REVIEW (60%)	MILESTONE 3 REVIEW (90%)
Location of Building, Pipe rack and Major Structures (Piping dummy models)	CSA	√		
Pipe (Preliminary route with actual size, DN300 & above)	M&BS	√		
Equipment (Preliminary size, position, orientation) including Equipment Support, Outline for future equipment	M	√		
Valves (Preliminary size and position, DN300 & above)	M&BS	√		
ACMV ducts (Preliminary size and route, width 600mm & above) and Cable tray routing (Main)	BS	√		
Preliminary Space allocation for Electrical cable trays (Main, preliminary size and route, width, 300mm, 600mm and above)	E	√		
Electrical Switchboards (HVSG, LVSG, MCC and MDB) (Preliminary size and location)	E	√		
LPS and earthing system (Preliminary routing, earth bar size and location)	E	√		
Equipment + LCS location with 600mm and above (Preliminary size and location)	E	√		
Main cables per tender cable schedule (Preliminary route and size)	E	√		
Downstream sub-DB (Preliminary size and details)	E		√	
Lighting and small power DB + lighting fixture (Preliminary size and details)	E		√	
All Equipment like JBs/Network process and non-process/ controller boxes etc. and Instrument (Preliminary size and position)	ICA		√	
Preliminary space allocation for Cable tray routing for 300mm, 600mm and above (Main)	ICA	√		



SUBJECT INPUT	Discipline	MILESTONE 1 REVIEW (30%)	MILESTONE 2 REVIEW (60%)	MILESTONE 3 REVIEW (90%)
Instrumentation cable trays (Preliminary route and size)	ICA		√	
Panels (Preliminary size and location with 600mm and above)	ICA	√		
Cast-in/Concrete slab/Wall openings (Size and location)	CSA&M&B S&E&ICA	√		
Access ways/Lifting path/Maintenance access/Operational access/Space consumers consideration	M&BS&E &ICA	√		
Escape Routes, Ergonomic Requirements	M&BS&E &ICA	√		
Underground Pipe	CSA & M	√		
Complete AMS Data drop 1 & 4D/5D codes	CSA&M&B S&E&ICA	√		
All comments in Milestone 1 addressed	CSA&M&B S&E&ICA		√	
Primary and Secondary Structures (Details)	CSA		√	
Ladders, stairs, railings and platforms	CSA	√	√	
Paving and Grading	CSA	√	√	
Pipe (DN80 and above) (Confirmed route and details with no clashes)	M&BS		√	
Major items as in approved P&IDs shown in model (include eye wash, safety shower location, except drainpipe, vents etc.).	M		√	
Major equipment (pumps, compressors etc.) sizing completed and shown in model, as per selected equipment dimensions	M		√	
Valves (DN80 and above) (Final)	M&BS		√	
ACMV ducts (width 300mm & above) and Cable Tray (Main & branch out, width 300mm & above), confirmed route and size and no clash	BS		√	
Firefighting systems (Fire and gas detectors, deluge system, fire extinguishers, fire monitors, hydrants)	BS		√	

SUBJECT INPUT	Discipline	MILESTONE 1 REVIEW (30%)	MILESTONE 2 REVIEW (60%)	MILESTONE 3 REVIEW (90%)
All remote mounted instruments in stanchion	BS		√	
Electrical cable trays with 300mm and above (Confirmed route and size, no clash)	E		√	
Electrical Switchboards (HVSG, LVSG, MCC and MDB) (Confirmed size and location)	E		√	
LPS and earthing system (Confirmed routing, earth bar size and location)	E		√	
Equipment + LCS location (Confirmed size and details)	E		√	
Main cables per tender cable schedule (Confirmed route and size)	E		√	
Downstream sub-DB (Confirmed size and details)	E		√	
Preliminary Lighting and small power DB + lighting fixture (Confirmed size and details)	E		√	
Instruments like flowmeter, switch, and transmitter (Preliminary size and positions)	ICA		√	
Tapping points (Final)	ICA		√	
Isolation valves (Final)	ICA		√	
Instrumentation Cable Tray c/w cables (Main & branch out Confirmed route and size, no clash)	ICA		√	
All remote-mounted instruments in stanchion	ICA		√	
Panels (Confirmed size and location)	ICA		√	
Supports for major items (Preliminary location and type)	M&BS&E & ICA		√	
Complete AMS Data Drop 1 & 2 and 4D/5D Codes	CSA&M&B S&E&ICA		√	
All comments addressed in Milestone 1 and 2	CSA&M&B S&E&ICA			√
Final dimensions of structures and steel constructions (Including pipe rack design)	CSA			√
Secondary steelwork and bracing	CSA			√



SUBJECT INPUT	Discipline	MILESTONE 1 REVIEW (30%)	MILESTONE 2 REVIEW (60%)	MILESTONE 3 REVIEW (90%)
All equipment (pumps, compressors, fans, cranes, panels, instruments, cable trays &, etc.) design approved and shown in the model, including minor items such as vents, drains, Eyewash, Safety shower, wash hose station, calibration column, drip trap, etc.	M			√
All pipes design approved and shown in the model, including vents, drains, temporary piping etc.	M			√
Pipe, equipment, and valves (Final complete with all details)	BS			√
ACMV ducts (Final with all details)	BS			√
Electrical cable trays (Final with all details)	E			√
Electrical Switchboards (HVSG, LVSG, MCC, and MDB) (Final with all details)	E			√
LPS and earthing system (Final with all details)	E			√
Equipment + LCS location (Final with all details)	E			√
Main cables per tender cable schedule (Final with all details)	E			√
Downstream sub-DB (Final with all details)	E			√
Lighting and small power DB + lighting fixture (Final with all details)	E			√
Instrumentation cable trays (complete including primary and branch out with supports)	ICA			√
Process Hook-up (all tubing)	ICA			√
Panels (Final with all details)	ICA			√
All elements established	ICA			√
Panels (Confirmed size and location)	ICA			√
Supports (Design confirmed with all details)	M&BS&E &ICA			√

SUBJECT INPUT	Discipline	MILESTONE 1 REVIEW (30%)	MILESTONE 2 REVIEW (60%)	MILESTONE 3 REVIEW (90%)
All clash cases are identified and cleared by S.O with “Accepted to proceed construction”.	CSA&M&B S&E&ICA			√
Details of all Interfaces within contract/other contracts are shown in the model and report.	CSA&M&B S&E&ICA			√
Complete AMS Data Drop 1 & 2 and 4D/5D Codes	CSA&M&B S&E&ICA			√

5) Report

For each module review, commenting shall be on a per discipline basis (Piping, structures, mechanical, electrical, construction, operation, and maintenance etc.) to achieve a more structured review.

After each review the team shall decide if the level of completion of the model is in line with project requirements. Any shortcomings shall be highlighted and after validation, it should be checked during the next review.

Comments will be tagged. Each tag shall be complete unambiguous and if necessary, includes a screen “snapshot” to avoid any confusion at the next stage of the review.

The area piping engineer shall coordinate the tagging and commenting process.

Minutes of meeting shall be written by Discipline Lead within 7 days after the review. The minutes of meeting shall be written in the form wherein the actual comments are presented with their tag number, description, date, and the discipline responsible for resolving the comment.

Actions will be tracked for closure complete with the level of documentation required to close the action.

3.1.7 Clash Detection and Resolution Workflow

In the preliminary design, detail design and construction stage, , the BIM team will update 3D models according to the design input firstly and then do the clash detection using BIM authoring tools like OBD, OPM. As shown in the following workflow, the (.i.dgn) federated models for different disciplines inside contract or (.i.dgn) federated models for different contracts will be combined to do clash detection, which make it possible to check the attributes in the clash detection results. Then the clash detection results can be generated from the BIM authoring tools. CES BIM Team and Engineer Team will work together to sort out the results following the classification standard in Table 10-2 Classification of Clashes and Clash report template Appendix A.7~A.8, 131242-WSW41-FM-IM-0007~0008. For the clash inside the contract, BIM team will arrange review with design team to resolve the clash. For the clash between contracts, BIM team will output the clash reports with the proposed solution (.xlsx). The clash reports will be discussed in weekly clash coordination meeting with S.O. with other related contracts. Then BIM team will update the model and clash report after the discussion and submit them in the monthly VCM. After all, highlighted clashes are resolved, then the related 2D drawing should be updated and resubmitted too. After that, the process can be ended. The tolerance is set no more than 1 mm in Bentley clash detection tools.

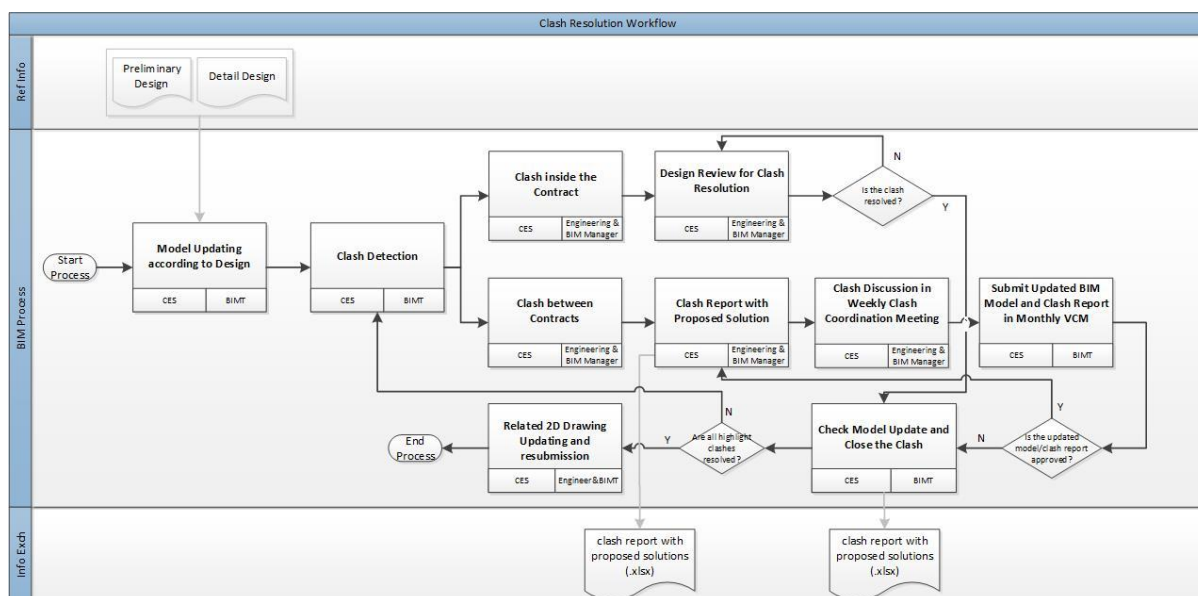


Figure 3-8 Clash Resolution Workflow

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.1.8 Model and P&ID Consistency Check Process

In the preliminary design and detail design stage, after the P&ID drawing and list report are approved by S.O., we will use the P&ID drawings to support the 3D model update. Then we will export the equipment, instrument, valve, pipeline list reports from the 3D model and check with P&ID manually to ensure that they are consistent. If they are not consistent, we will develop the 3D model to match the information of P&ID. Only when they are all consistent, then the process be ended. The below table is the reference for the consistency check report.

Table 3-1 Consistency check report list

TAG Number	Description	P&ID Drawing Number	BIM(3D) Model Number	Reason for inconsistency

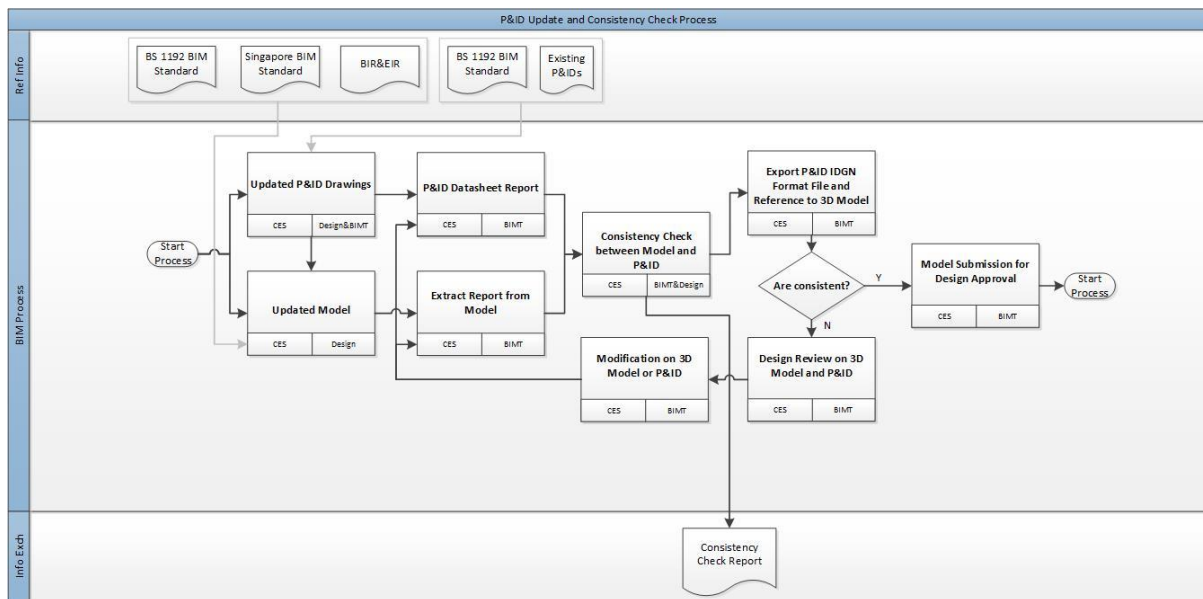


Figure 3-9 P&ID Update and Consistency Check Process

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.2 Construction Coordination

In the construction stage, construction model will be used for construction coordination. General procedure for construction coordination is shown as Figure 3-10.

- 1. Proper Models Placement:** All parties to ensure that latest model help been completed, and all models have the same coordinate system and are accurately aligned to each other to avoid errors during coordination.
- 2. Combine Models:** BIM team to combine models from different disciplines, facilities or even packages, with selected coordination tool. .i.dgn format are suggested if the models are created and updated with different BIM authoring tools.
- 3. Specify Scope of Model Checking:** Models can be divided into several zones based on the process system for model checking (See Section 3.1.6 2)). BIM team to discuss with engineer team on the checking items of different disciplines for each zone.
- 4. Determine Rules for Model Checking:** BIM manager and Discipline leads to give general rules for each checking items before coordination works start. The rules should include both design rules and modelling rules.
- 5. Carry Out Model Checking:** Use the coordination software to run crash detection and identify collections and design errors.
- 6. Assigning and Communicating Issues:** Assign detected issues to responsible team members and communicate the need for resolution.
- 7. Send Coordination Reports:** Generate and send coordination reports to team members, highlighting the identified clashes and issues.

8. Results Analysis and Further Steps: Analyse the coordination reports to identify patterns, bottlenecks, and areas for improvement, and take appropriate actions to address the identified issues.

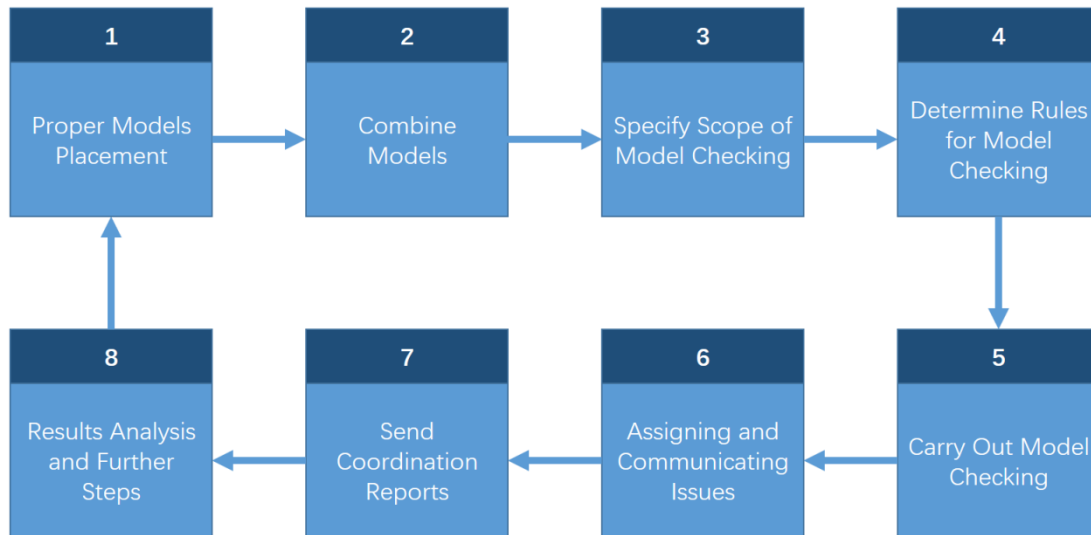


Figure 3-10 General Procedure of Construction Coordination

The general procedure above, detailed items below can be done for construction coordination.

1. The inspected completed works will be updated into native model in one week if big changes have been made on site. BIM team and Engineer team can review and find issues in time if they influence related uninstalled works.
2. The model will be updated with TWRP Information and shared among all interfacing contractors monthly. Contractors will get the latest construction status information of other contractors in the model.
3. 4D codes will be input into native models. 4D simulation work will be done to let related parties know the construction schedule visually.
4. Clash detection should be done once a month based on the latest construction model. The clashes should be recorded in clash reports. Each items in clash report shall be lodged with the appropriate person-in-charge, date, and priority and distributed to all parties for recording and follow-up action purposes.
5. BIM team will extract accurate material take-off from the detailed construction model and help project team arrange the purchase and fabrication work on site.
6. Regularly scheduled Design Review Coordination Meetings will be conducted to ensure construction coordination in time.

BIM team will try to fully utilize BIM tools and help project teams coordinate construction work more efficiently.

3.3 Health and Safety

In the preliminary design, detail design, and construction stage, as shown in the following workflow, HSE Manager, Engineering Manager, and BIM manager will work together to define whether there are any unhealthy or unsafe factors before construction through reviewing the 3D model and 4D simulation. Based on the review, project team will give solutions to these unhealthy or unsafe factors. Then these factors will be tracked and documented until these unhealthy and unsafe factors are completely solved.

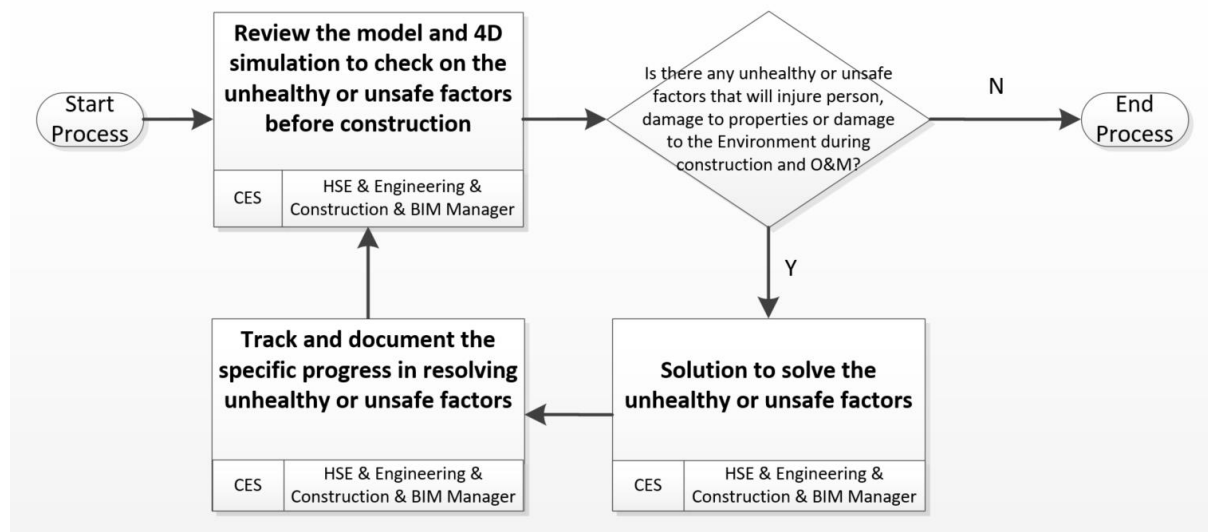


Figure 3-11 Health and safety check through BIM

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

3.4 4D Construction

4D construction model will be done with embedded programme for the Works. The model shall include temporary works, planned permitting works, and interfaces with any other contractors as a minimum. In the design stage, 4D simulation works will be done based on the planned schedule and used to optimize the design considering the construction sequence. In construction stage, 4D simulation will be done before construction on site to assist construction team to arrange the works.

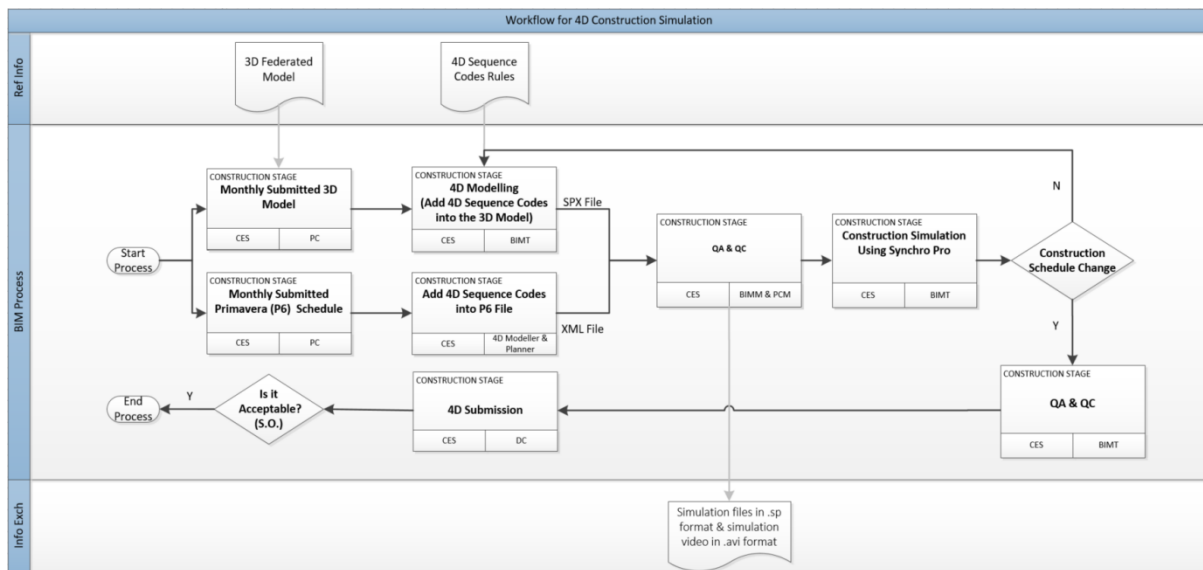


Figure 3-12 Workflow for 4D Construction Simulation

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

The 4D works should be based on the monthly VCM model and construction schedule. Firstly, the contractor should define the 4D coding rules of Mechanical, Electrical, ICA, Structure, and Architecture. After the 4D coding rules are approved and the referenced information of the P6 schedule is completed, 4D modeler should add 4D codes into the 3D model and project planner should add 4D codes to the P6 schedule. Both of these should be checked by BIMM and PCM. Then with the input of the P6 and 3D model, Synchro Pro can match the P6 schedule and 3D model through the 4D codes and simulate the 4D sequence automatically. Once the construction schedule is changed, the 4D simulation needs to be adjusted accordingly. The 4D work should be reviewed by PCM, Quality Manager, and BIM manager before submission, and the review work should follow the checklist 131242-WSW41-FM-IM-0004, and then output the simulation files (.sp) & simulation video (.avi). The 4D outputs' complete date and submission date should all be six months ahead of the construction. The 4D native model and simulation video shall be submitted every 8th of the month and included in the monthly VCM submission.

3.5 5D

5D model will be done with associated exports of data from the model. In design stage, the material list will be extracted from the model with BoQ codes for the engineering team to evaluate the cost before construction on site. In the construction stage, 5D reports will be extracted from the model for those completed works on-site for project expenditure tracking and verification of monthly progress claims.

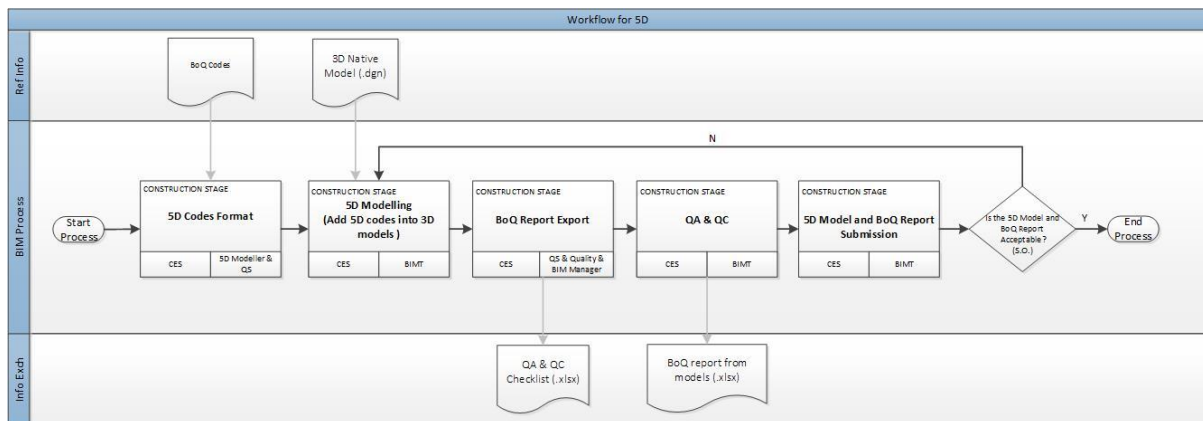


Figure 3-13 Workflow for 5D Cost Evaluation

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

Firstly, 5D modeller and QS will define 5D codes based on BoQ codes. Then 5D modeller will input the 5D codes into models and export the 5D reports from models. The 5D work shall be reviewed by QS, Quality Manager, and BIM manager before submission, and the review work should follow the checklist 131242-WSW41-FM-IM-0005, and then output the BoQ report from models for project expenditure.

The 5D model submitted will contain 5D codes and claim status and claim date information when it is available. The BoQ report shall be submitted every 8th of the month included in the monthly VCM submission.

3.6 As-Built and Asset Management Data

During all stages of C4B projects, the collection and management of 6D asset data will be carried out and should be divided into three parts, including datadrop 1, datadrop 2 and datadrop 3. Datadrop 1 should be collected and managed during the detailed design stage, including equipment requisition, all details from the final approved data sheet, etc. Datadrop 2 should be collected and managed during delivery, construction, and installation stage, including installation details, FAT test sheets, 4D codes, 5D codes, etc. Datadrop 3 should be collected and managed during the commissioning, start-up, and operation stage, including maintenance task details, SAT test sheets, O&M training & instructions, HAC register, confined space register, etc. These three parts of asset data should be determined according to the specific requirements of PUB and S.O. They will be accumulated in-depth during the project construction stage and approved in stages, ultimately achieving comprehensive asset delivery.

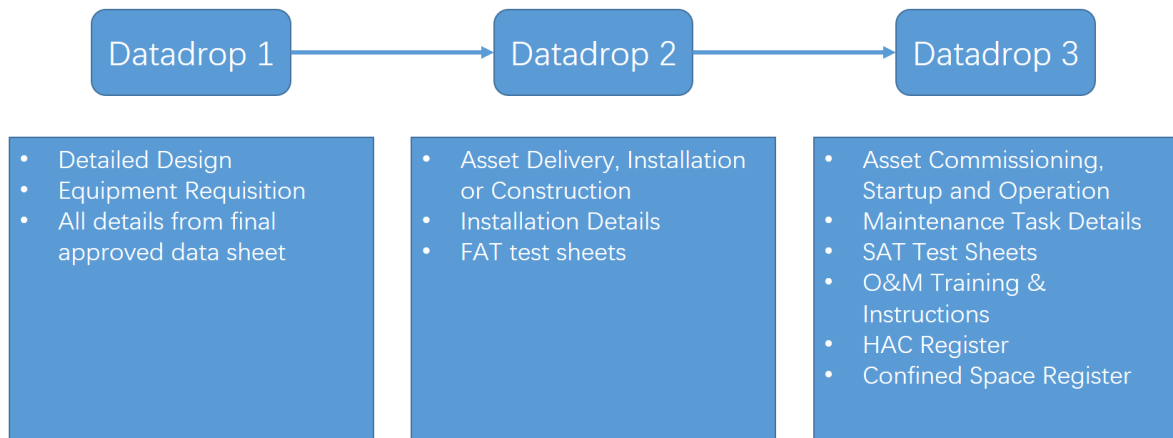


Figure 3-14 6D Asset Data Requirement

4 Model Requirement

4.1 Modelling Approach

The BIM work, including 2D Drawing Extract, P&ID Drawing Development, 3D Model Development, 4D Simulation, 5D Report Export, 6D Asset Information Management, should be carried out by the authoring tools shown in below. The software version is as per requirements.

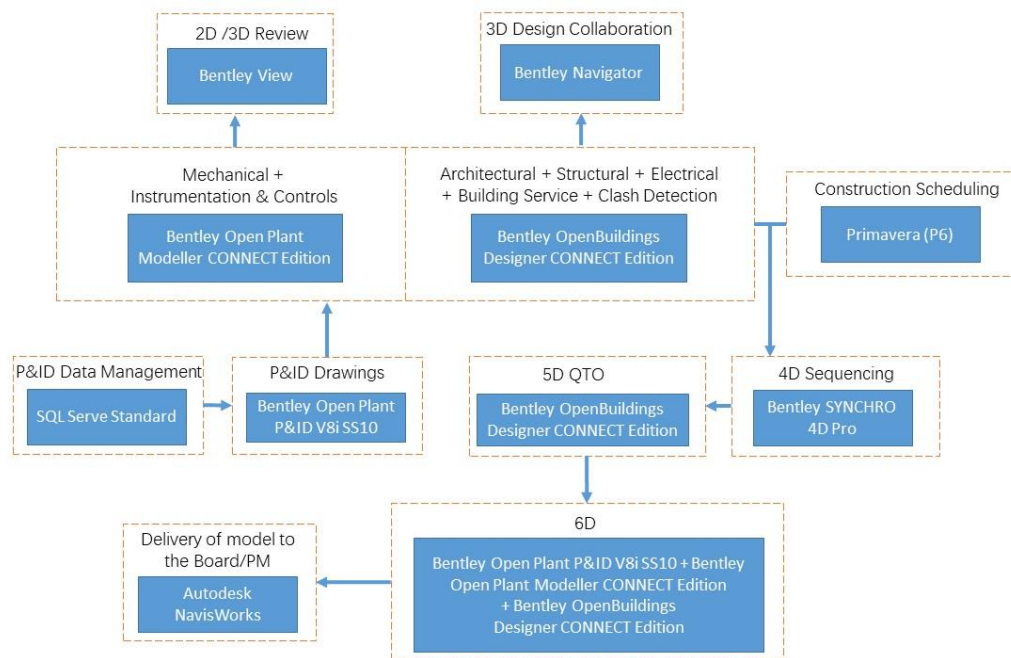


Figure 4-1 Authoring Tools for BIM Works

For Civil, Structure, Architecture and Electrical discipline modelling work, it should be done by OBD. For Mechanical and Instrument Control Automation discipline, the elements shown in P&ID drawings like the equipment, pipes, valves, and instrument should be modelled by OPM and others should be modelled by OBD.

Especially for 2D drawing production work, it will be carried out using OBD (For model created by OBD) and OPM (For model created by OPM), following the procedure as shown below. A blank .dgn model will be created by authoring tools and work as a Dynamic View (DV) Container. Then drawing generation-related model will be referenced into the DV Container and the layout/section drawing models will be cut in it. The DV Container realizes dynamic connection between model and drawing, and future model changes will be updated into the layout/section drawing models directly. Then the drawing model will be referenced into drawing sheet to generate the 2D drawing. Finally, 3D models and 2D drawings will be submitted together for review.

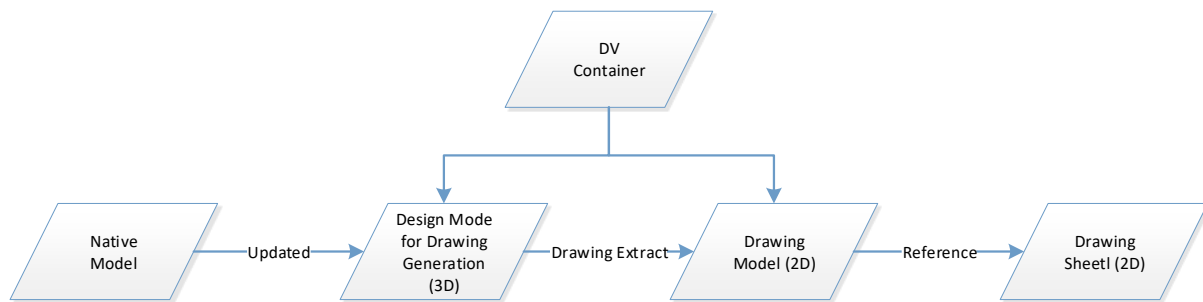


Figure 4-2 2D Drawing Generation Procedure

Especially for 3D modelling production work, it will be carried out using OBD for architectural, structural, electrical, ICA (except instruments), building service models and OPM for mechanical models, following the procedure as shown below. The native model will be updated based on design input. These model update work must be done by using OBD or OPM with official issued TWRP software workspace. Then the last native models will be referenced into each discipline federate model. Then each discipline federate model will be referenced into the corresponding facility model. Finally, each facility model will be referenced into the overall model.

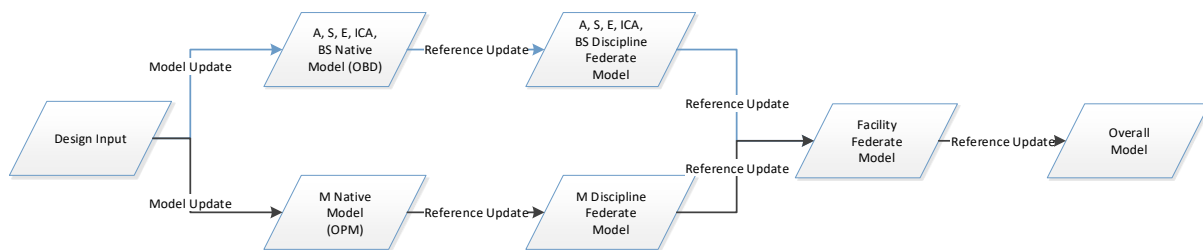


Figure 4-3 3D Model Production Procedure

4.2 Level of Detail

With the Level of Development per Milestone in EIR, we developed our Level of Detail and Level of Information for the models. The Level of Detail can be referred to LOD matrix instructions (Appendix B.1-131242-WSW41-RG-IM-0001). Below is the Level of Information and an example of equipment from LOD-Spec-2021 for reference.

LOD	Description	Example	
100	Diagrammatic or schematic model elements; Conceptual and/or schematic layout.	NA	
200	Schematic layout with approximate size, shape, and location of equipment; Approximate access/code clearance requirements modelled.	 <i>126 D2030.10-LOD-200 Stormwater Drainage Equipment</i>	Schematic layout with approximate size, shape, and location of equipment.

300	The quantity, size, shape, location, and orientation of the element as designed can be measured directly from the model without referring to non-modelled information such as notes or dimension call-outs. The project origin is defined, and the element is located accurately with respect to the project origin.	 <i>127 D2030, 10-LOD-300 Stormwater Drainage Equipment</i>	Access/code clearance requirements modelled.
400	An LOD 400 element is modelled at sufficient detail and accuracy for fabrication of the represented component. The quantity, size, shape, location, and orientation of the element as designed can be measured directly from the model without referring to non-modelled information such as notes or dimension call-outs.	 <i>129 D2030, 10-LOD-400 Stormwater Drainage Equipment</i>	Actual access/code clearance requirements modelled.
500	LOD 500 relates to field verification and is not an indication of progression to a higher level of model element geometry or non-graphic information.	NA	

Table 4-1: Level of Development Per Milestone

Project Milestone	Conceptual Design/Feasibility Study	Preliminary Design	Detailed Design	Construction	As-built
LOD	LOD 100	LOD 200	LOD 300	LOD 400	LOD 500

4.3 Model Coordination System

4.3.1 Key Aspects of Model Coordination System

BIM model coordination system involves integrating and aligning various discipline-specific models, such as architectural, structural, mechanical, electrical, and plumbing models, into a centralized BIM environment. This coordination ensures that all elements fit together seamlessly, avoiding clashes and potential conflicts that could arise during construction. The approach is not to build a single model of the entire project. Instead, multiple models (with associated data) will be generated and

referenced together into a Federated Model. The details of the model structure can be referred to Appendix B.2-131242-WSW41-RG-IM-0002.

Below are some key aspects in the BIM model coordination system:

Item	Description
Clash Detection	Detect clashes between different building elements, such as beams intersecting with ducts or pipes conflicting with walls. By identifying these clashes early on, the project team can resolve them before construction begins, reducing errors and rework during the construction phase.
Collaboration	Facilitates real-time collaboration among project team members by working on a shared model. Foster the communication and ensure that everyone is working with the latest and most accurate information through CDE.
Version Control	Include version control mechanisms, allowing team members to track changes and revert to previous versions if needed.
Model Federating	Different disciplines create individual BIM models that need to be federated into a single, comprehensive model to visualize the entire project. The federated model provides a holistic view of the building and helps to identify potential issues.
Data Sharing	Enable the exchange of data between different software applications, enhancing interoperability and data continuity throughout the project.
Issue Tracking	Provide tools for issue tracking and resolution. Team members can document clashes or coordination issues, assign responsibility, and track the resolution status.

4.3.2 Model Structure and File Naming Convention

To ensure consistency across all contracts, the BIM Team will comply with a common Model File Naming Convention. Model File Naming Convention contains five-part: contract code, project prefix, document type, discipline, and description/ Sequence. Here is an example of how to use file naming conventions on BIM models.

Table 4-2 Example for model file naming

Contract Code	Project Prefix	Document Type	Discipline	Description/ Sequence
131242	WST02	M3	M	8600

As shown in Table 4-2, the C4B project's contract code is 131242. The project is separated into 3 areas (WST02, WSW41, WSD11 which is served as the project prefix). M2, M3, M3, M4, M5, M6, CM, CR is the abbreviation of the different document types. A, S, M, B, E, I, P is the abbreviation of different disciplines. The last field in the Model File Naming Convention is a sequence range from

8900 to 8999. For the version codes, it will use A.1, A.2, ... for minor revision and A, B, ... for major change to track the model changes in construction stage. Details can refer to 131101-WGS00-RG-IM-0008.

4.4 Model Accuracy

4.4.1 Model Geometry Accuracy

The BIM model should accurately represent the physical dimensions and shapes of the building elements, such as walls, floors, columns, doors, and windows. Precise geometric accuracy is crucial to avoid clashes and conflicts during construction. Therefore, BIM models from different disciplines (architectural, structural, mechanical, electrical, plumbing, etc.) and contractors must be accurately coordinated to avoid clashes between various building systems and elements. This requires all parties to do frequent visual inspection and measurement inspection during work and strict dimensional and spatial position inspection before publishing to other parties. The procedure is shown in Figure 4-4. First, BIM Modelling Team update model based on the latest design input. Second, BIM authoring lead will check the size and LOD of the model to ensure the size is consistent with the design input and the LOD can meet the requirement of project use. Third each discipline lead will review the model and ensure the size meets design requirements. Then CES will conduct a BIM walk through to involve all related project members to review the model. Key size checks can be done in the walkthrough and related comments on the model geometry will be recorded for the next BIM update.

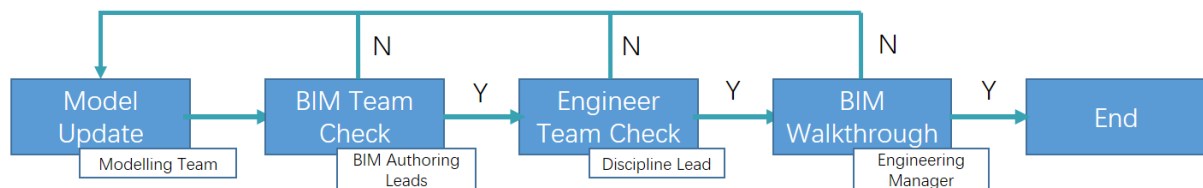


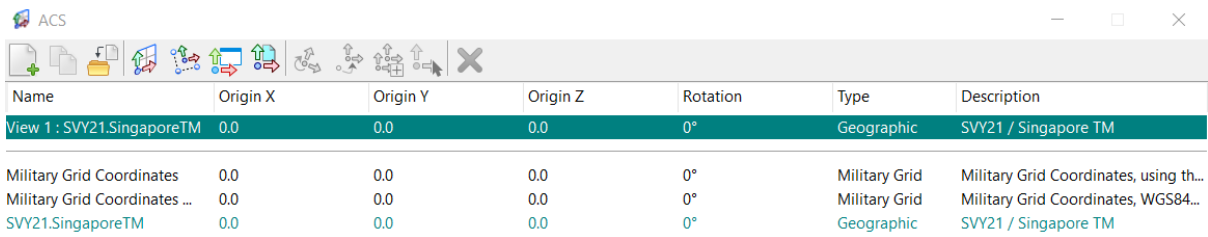
Figure 4-4 Model Geometry Check Procedure

4.4.2 Model Coordinates Accuracy

To ensure the accuracy of elements spatial location, the BIM model shall be created using the real-world coordinates and shall comply with the Land Surveyors Board (LSB) Directive on Engineering and Hydrographic Survey Practices and the Standard and Specifications for 3D Topographic Surveying (Mapping) in Singapore (Version 1.0 – 2013) produced by the Singapore Land Authority (SLA). All project spatial information shall be referenced to the Singapore Transverse Mercator Grid (SVY 21 datum – 2006), with vertical control derived from the SGeoid09 - geometric geoid model established by SLA. Any GNSS survey work shall take reference to the Singapore Satellite Positioning Reference Network (SiReNT). To ensure that models are correctly/properly georeferenced, we should check whether the auxiliary coordinate system of the models is referenced with Singapore Transverse Mercator Grid (SVY 21 datum – 2006) (SVY21. Singapore TM) as shown in the following table and picture.

Table 4-3: Project Coordinate System

Projection	SINGAPORE TRANSVERSE MERCATOR
Grid zone	TBC
Spheroid	WGS1984/SG09
Plan Datum	PM
Height Datum	SVY21
Unit	SINGAPORE HIGH DATUM (SHD)



Name	Origin X	Origin Y	Origin Z	Rotation	Type	Description
View 1 : SVY21.SingaporeTM	0.0	0.0	0.0	0°	Geographic	SVY21 / Singapore TM
Military Grid Coordinates	0.0	0.0	0.0	0°	Military Grid	Military Grid Coordinates, using th...
Military Grid Coordinates ...	0.0	0.0	0.0	0°	Military Grid	Military Grid Coordinates, WGS84...
SVY21.SingaporeTM	0.0	0.0	0.0	0°	Geographic	SVY21 / Singapore TM

Figure 4-5 Auxiliary coordinate system referenced with Singapore Transverse Mercator Grid (SVY 21 datum – 2006) in model.



Figure 4-5 TOPOGRAPHICAL SURVEY PLAN

Table 4-4 CONTRACT BOUNDARY CO-ORDINATES

NO	NORTHING	EASTING
B1	29499.72	6727.79
B2	29523.93	6729.66
B3	29549.86	6741.30
B4	29552.97	6751.92

B5	29547.63	6821.70
B6	29544.31	6828.14
B7	29536.75	6831.58
B8	29491.95	6828.20
B9	29488.78	6814.21
B10	29443.17	6810.69
B11	29447.38	6756.20
B12	29493.48	6759.82

4.4.3 Model Data Accuracy

The information contained within the BIM model should be based on reliable and up-to-date data from CES engineer team, subcontractors, suppliers, etc. This includes material specifications, product details, construction information, and any other data relevant to the project. Inaccurate data can lead to incorrect assumptions and decisions during the construction process. So, during work, all project stakeholders need to do effective communication and collaboration to ensure that accurate information is shared and integrated into the BIM model.

To ensure the model data accuracy, the contractor should export the datum of all elements from model every month or after model update. The datum will be checked by CES engineer team and BIM team. The engineering team (Discipline lead) mainly check if the datum content is correct and latest. BIM team (BIM manager and BIM authoring lead) mainly check the datum format. In milestone model submission, CES BIM team will export data with required report content and format from the model and submit the report and model together for S.O. review. Monthly CES BIM team will submit VCM files with information needed for S.O. to check.

4.4.4 Attention Points

To ensure the model accuracy, the items below need to pay more attention.

- a) The common modelling system of measurement will be a metric. Models will use consistent units across the project and project units will be millimetres of integer to achieve a sufficient level of accuracy.
- b) The final product in the model including actual finishes, model, size, and location of fixtures, design-intent modulation of tiles and ceiling, type and location of furniture, etc.
- c) Interoperability should ensure that the data created by each party translates correctly in terms of both geometric and non-geometric attributes for accurate reference by all users of each model.
- d) Objects from different disciplines need to be coordinated to reference points.

- e) 3D model information should be in accordance with P&ID data, the consistency should be checked automatically through OpenPlant Modeller and manually by the BIM team.
- f) Quality management is required as listed in Section 6.2 to ensure the model's accuracy.
- g) The BIM Authoring tools for each discipline can be referred to in Table 8-1.

5 Collaboration

5.1 Common Data Environment

5.1.1 Purpose of CDE

Common Data Environment (CDE) is a common digital space that hosts relevant information for collaboration, exchange, and communication to deliver a project.

The CDE aims to achieve the following goals:

- Ensure consistent and better-managed project information and process for projects.
- Allow project team members timely access to up-to-date, relevant, and reliable project information in a common and structured environment for the delivery of a project.
- Facilitate close collaboration among project team members through sharing, exchanging, communicating, and managing the project information in a common space.
- Enable project teams to derive useful project performance insights for trending purpose.

5.1.2 CDE Settings

In this project, ProjectWise is used to serve as contract's own CDE platform for BIM, which will serve as a single source of information during works, and to meet the security & backup requirements for BIM data. The Server Settings and ProjectWise Settings are shown below. The Main contractor, Subcontractor (if required), and Supply chain (if required) will access to ProjectWise via VPN. For interface management, the interface data to & from related contracts will be shared and received after approval by S.O. via Prolog.

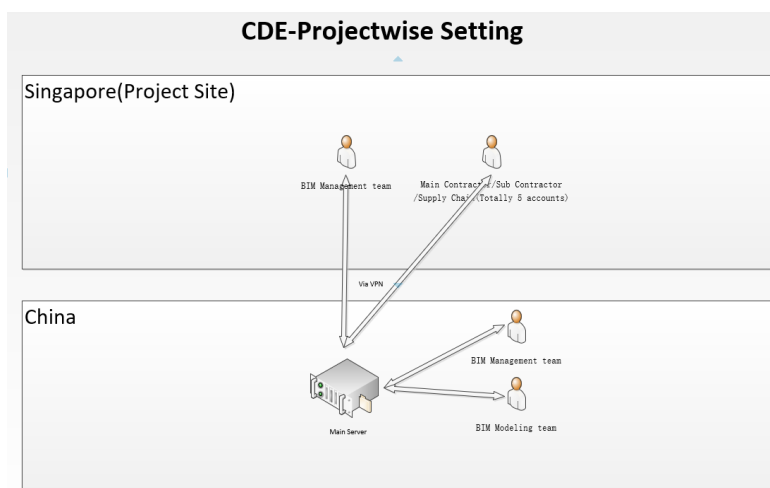


Figure 5-1 The ProjectWise Server Setting for BIM CDE

5.1.3 CDE Folders Structure

The folder structure of the CDE is established as defined in BS 1192:2007 work processes, this workflow for CDE served as a central point of all project in-formation as set out in PAS 1192standards. The details of each folder are shown in below Figure 5-2 and Table 5-1.

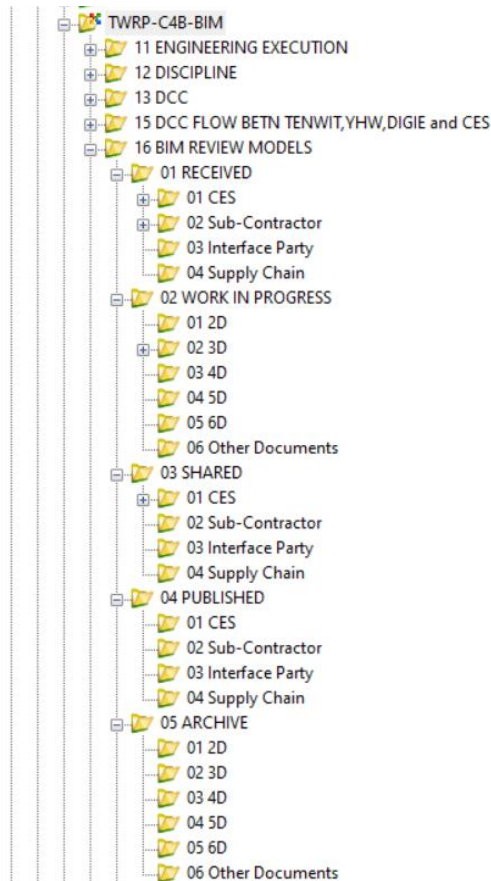


Figure 5-2 Snapshot of Actual Folder Structure in ProjectWise

Table 5-1 Details for CDE Data Container

Folder Structure	Content	Description
Received	Drawings, Models, Comments Document, Interface Document, etc.	“Received” folder is used to store the raw information from each party. Data will transfer to the Work In Progress workspace after check by BIM Manager.
Work in Progress	Non-approved progress data	“Work in Process” area of the CDE is where project teams (disciplines) carry out their work. Prior to moving BIM information to the “Shared” location on the project team’s CDE, BIM Authoring Lead shall perform a series of QC checks to verify their BIM models meet the requirements. Once

		approved by the BIM Authoring Leads, each team is required to post their BIM information into the Shared.
Shared	Non-authorised data for inside contract share	“Shared” area of the project team’s CDE will be reserved for BIM information that is to be considered approved by BIM Authoring Leads and fit for coordination. This means other disciplines can reference this information for ongoing design development.
Published	Authorised data for public share	All BIM deliverables will be checked and reviewed by the BIM Manager to ensure the BIM deliverables fully meet the requirements. Once approved by the BIM Manager, the BIM information can be posted to the “Published” area. “Published Documents” are to be considered output deliverables for the total project team. This is production information suitable for Tender or Construction.
Archive	Verified data	Project history maintained for knowledge and regulatory and legal requirements. As-Built drawings. As constructed models. Change audits. Asset data. Operation and maintenance. Health and Safety

For different containers, the BIM team have already a default Document setting as below figure, which will take each kind of data for different parties into consideration.

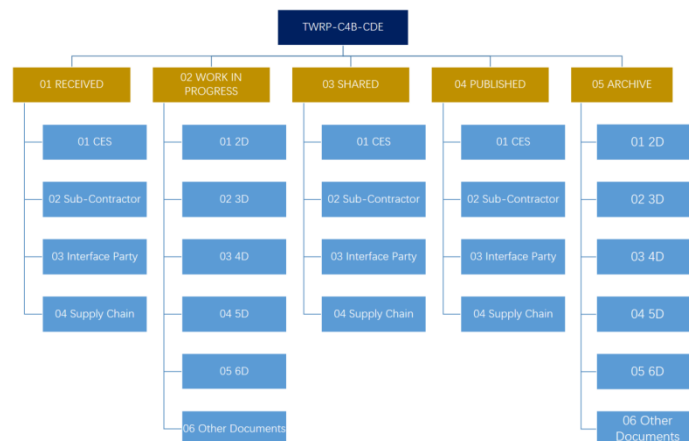


Figure 5-3 Common project folder structure

5.1.4 Data Exchange Procedure in CDE

One of the key functions of the CDE is to facilitate real-time sharing of project information among all project stakeholders in a collaborative environment, thus improving communication and coordination efficiency. Because of this intention, the contents and recommendations of the following sections should be implemented before the start of the project to promote the trend of project information sharing. The data exchange procedure is shown below.

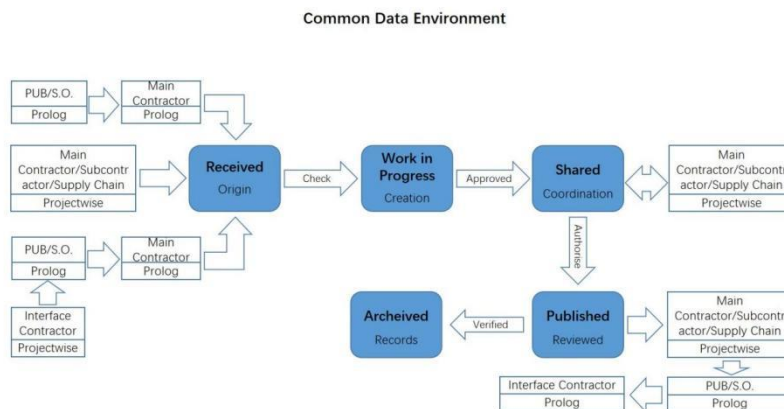


Figure 5-4 Typical Workflow for CDE

For document control, CES BIM Team also establishes a matrix for management of all documentation transfers between the CES and the S.O

Table 5-2 Document Transmission Distribution Matrix

	Prepare	Approve
Receiving	Document Controller	BIM Manager
Distributing	Document Controller	BIM Manager

5.1.5 CDE Data Safety

To ensure the safety of the data, the contractor will provide Multi-Tier Cloud Security to protect the data against cyber-attacks. The structure of the security system is shown in Figure 8-2 below. In the Group Level contractor will have Security Protection System to protect project data against the intrusion from the internet. At Company Level there will be tools like firewall, WAF, IPS, Gate Keeper to ensure data safety. Project members are required to use VPN to access project data. At Project Level, contractor will have a specialist to manage the member access rights and backup project data monthly. At User Level, the system will record the user's behaviour on data operation and the team will provide training to specify the key points of data protection.

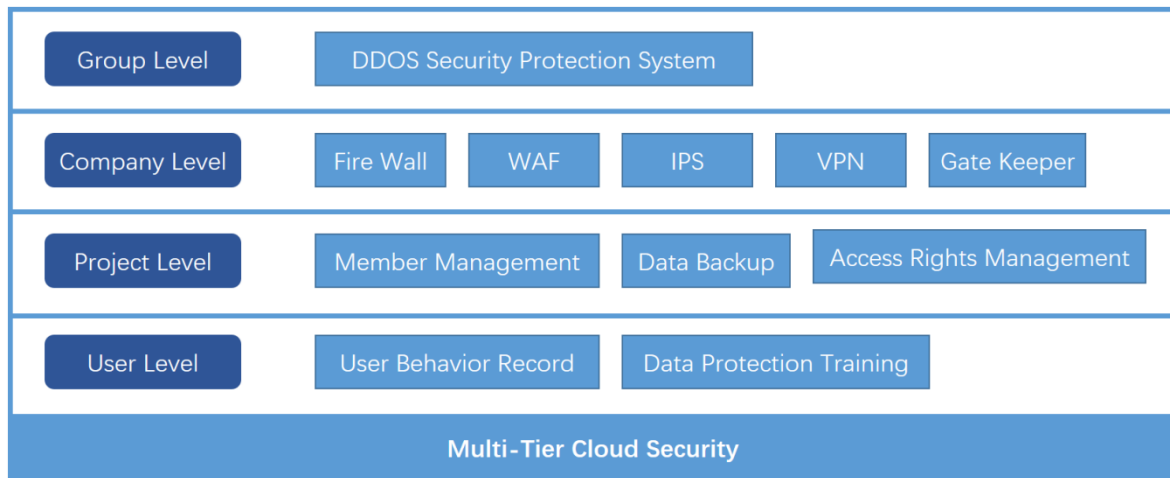


Figure 5-5 Multi-Tier Cloud Security

5.2 Communication and Coordination Meeting

5.2.1 BIM Document Correspondence through Prolog

All BIM document correspondence with S.O., including document submissions, questions, and replies, will be delivered through Prolog. CES will submit VCM through Prolog for S.O. review on the 8th of each month. The monthly VCM will include below 6 items:

Table 5-3 VCM submission items

Item	Content	Format	Mark
1	Native Model	.dgn	Submit in January and July every year.
2	Federate Model	.i.dgn, .i.bim , .nwd	Facility model and overall model.
3	Clash Report	.xlsx	
4	4D	.sp, .avi, .xer	Submit 6 months before construction.
5	QA&QC	.xlsx	
6	Change Log	.xlsx	
7	MIDP	.xlsx	
8	Model Tree	.xlsx	
9	Comments Sheet	.doc &.PDF	Last month comments sheet

5.2.2 Data Change Request

During the BIM execution, some data may be missing in the OPM/OBD workspace or OPPID dataset. CES will summarize and clarify the data request base on the latest version workspace OPM/OBD workspace or OPPID dataset/database. The submitted data change request should follow the template as Appendix A.11 - 131242-WSW41-FM-IM-0011. After S.O. reviews and approves the request, a new version workspace or dataset will be updated to contractors.

5.2.3 Coordination Meeting

Exchanged files will be verified by the file name as part of the CDE upload or receipt process. Only correctly named files will be accepted. Correctly named files will be uploaded/accepted by the CDE without further profiling into the correct folder structure position.

Every week, BIM team will publish the Federated model and share between disciplines/trades, subcontractors and other parties in the supply chain . Federated models shall be shared and published on monthly basis to S.O.

Published files will be validated against the model and information requirement before being exchanged.

1) BIM progress meeting with S.O.

CES BIM Team will hold BIM progress meetings with S.O. bi-weekly. This meeting shall be attended by our BIM Manager, BIM Authoring leads and BIM Manager, BIM Coordinator of S.O., and any other meeting discussion related personnel as required.

The agenda at the meeting to include the following as a minimum:

- a) Introduce the last two weeks' progress in 2D, 3D, 4D, 5D, 6D works.
- b) Review S.O. comments of the submitted monthly model files.
- c) Status of the CES's BIM deliverables.
- d) CES's clash detection reports; and.
- e) Some other issues need to be discussed.

2) Interface meeting

Before meeting, C4B BIM team will combine the latest model with other contracts' VCM model which is shared by S.O. monthly. In the combined model, CES BIM Team will record the found issues and views for meeting discussion. Then all results in the meeting discussion should be recorded and tracked in the next VCM model.

Interface meeting will invite all interfacing contractors, C4B BIM team, CES interface manager and CES Engineers to attend the workshop .Interface manager will chair, control and minute regularly Interface Coordination Meetings and to promote cross-package interaction and close cooperation between all parties on the project. BIM team will drive model in the meeting and will updated model according to the MOM after meeting.

3) Internal BIM coordination meeting

For internal BIM coordination work between the BIM Team and Engineering Team, The BIM manager will get BIM input from Engineering Team and assign tasks to the BIM team. Then, according to the BIM work, the BIM team will coordinate with the corresponding department of Engineering Team.

And BIM team will also coordinate with S.O. directly for C4B BIM issues if required. All the BIM outputs that need QA/QC should be reviewed by Engineering, Quality Manager, PCM, and QS. All submission of BIM outputs should be managed by the document controller.

Internal BIM coordination meetings will be held weekly. This meeting will invite all Parties within the C4B project, including the design team, construction team, subcontractors, etc. The specific date depends on the current work. The model will be shared with all internal parties weekly before the meeting. The agenda of the meeting includes the following as a minimum:

- a) Clash detection report.
- b) P&ID updated.
- c) 3D Model Development.
- d) QA/QC.

4) Internal discipline issue discussion meeting

Internal discipline issue discussion meetings will be held weekly. The meeting will focus on resolving issues inside the discipline. This meeting attendees will include C4B BIM team and corresponding discipline designers. Before this meeting, the BIM team and the design team will raise discipline related issue for discussion and record the results in the meeting. Also, the progress of each discipline will be tracked in this meeting.

6 Compliance plan and Quality management

6.1 Compliance plan

6.1.1 BIM Protocols and Execution

The following table shows the BIM protocols used to create, manage, and deliver models and information to meet the EIR and contractor's plan to follow these protocols.

Table 6-1 BIM Protocols and Execution

No.	BIM Protocol	Execution
1	All models created should be both complete and accurate, in accordance with EIR and with the approved BIM Execution Plan.	Keep regular BIM walkthrough internally and externally to ensure the model include all necessary elements and ensure enough model accuracy for project use. The spatial information will be referenced to the Singapore Transverse Mercator Grid (SVY 21 datum – 2006).
2	The Information Model will be the sole source from which the following deliverables are to be derived:NA	
2.1	2D denotes drawings	All drawing sheets will be provided the soft copies for S.O. review/comment. All drawings will be generated using the BIM authoring tools and be reviewed on the design items by the engineering team and on the BIM standard by BIM team according to TWRP BIM/CAD Graphics Standard.
2.2	3D models created using the BIM Authoring tools.	CES will provide the BIM Models in both native file formats (.dgn), imodels (.i.dgn and .i.bim). CES BIM Team will give training to all parties, including C4B engineer team, subcontractors, suppliers, to ensure the full use of BIM authoring tools.
2.3	4D denotes construction sequence/program information	CES BIM Team will map 3D objects to the date/time as shown in the monthly P6 Programme. CES BIM Team will enable the visualization and simulation of the construction process. Target to provide aids in

		identifying potential construction issues, optimizing workflows, and improving overall project scheduling and efficiency.
2.4	5D denotes quantification and project expenditure tracking	The model with quantification schedules will be provided. CES BIM Team will integrate cost-related information with the BIM model to have better cost control and financial management throughout the project.
2.5	6D denotes asset management.	Prior to completion of construction, the CES will embed COBie and PUB's asset management data on all model elements that represent "Managed Assets". CES will check the datum format to ensure they are entirely suitable for incorporation into the Board's AMS.
3	A set of modelling guidelines for BIM elements at different stages of the project are provided in the latest version of the Singapore BIM Guide and shall be referenced. The BIM guidelines outlined in the Singapore BIM Guide are designed to promote consistency, interoperability, and efficient collaboration among project stakeholders.	CES BIM Team will reference the Singapore BIM Guide to improve BIM work including Element Naming and Coding, LOD, Geometry and Visualization, Data and Information Requirements, Coordination and Clash Detection, and Information Delivery and Exchange.
4	The Contractor shall maintain and update the Handover model(s) as the model author once the S.O. furnish the Handover Model	Based on the Handover model (if have), CES design team will further improve the design, and BIM team will refer to the standard model data and format to further update the model. And CES will collaborate with the S.O. and other project stakeholders together and incorporate relevant data and information into the model, to ensure that it accurately represents the final state of the constructed project.
5	The Contractor shall develop the Handover Model and manage the PIM by the MIDP within Contractor's CDE, and share it with S.O. for periodic review and approval	CES will be responsible for creating and developing the original model and should adhere to the master information delivery schedule specified in the MIDP. This BEP outlines the information deliverables and milestones throughout the project lifecycle. CES will ensure that the project information is collected and delivered by the schedule in MIDP.
6	The Contractor shall use the outputs from BIM models for 2D and 3D drawing production, visualizations, simulation, project expenditure analysis,	Non-geometrical data including schedule/datasheet describing the physical, functional, and performance of the plant and equipment will be provided. For example, OBD model data and TWRP information data

	collaboration, planning, and documentation throughout the project life cycle. These outputs shall include reports, extracted CAD Drawings, and non-geometrical data	of all models will be input into model. These data will be submitted together with model to S.O.. Especially, the data of elements in P&ID will be input into P&ID database. CES BIM team will export data report from the database to submit with P&ID drawings for S.O. review.
7	In general, each element shall be modelled according to its size, shape, location, orientation, and quantity. The Contractor shall develop each element with accuracy as the project progresses until the level of accuracy is sufficient to support construction and installation	BIM manager will improve the LOD matrix according to construction/installation requirements and based on BIMForum_LOD-Spec-2019. BIM authoring to check the model accuracy during modelling work. Details can refer to Section 4.4.1.
8	Drawings shall fully be derived and generated from BIM Models possible. Drawings shall include, but not be limited to plan and sectional views, elevations, profiles, quantities, and schedules; and shall be delivered as representations of or extractions from BIM Models	CES will ensure proper model formatting, annotation management, and clash detection, as well as detailed supplementation of drawings and version control to ensure consistency between drawings and models. Additionally, contractor will submit the model drawings for review to ensure design quality and improve review efficiency.
8.1	2D Drawings (typical details, notes, legends, etc.) not extracted from the BIM Models shall be identified in the Drawings List that all Project Teams will use to plan their document sets, and the use of these drawings will be supervised by S.O.	CES will make drawings register to record both model-generated drawings and non-model generated drawings. All disciplines will be required to update the latest drawings information, like drawing number, drawings content, drawing schedule, drawing status, etc, into the drawings register. All discipline will follow the schedule to carry out drawing work. The use of these 2D Drawings on the project shall be subject to the PM/S.O. review and approval prior to the creation of these 2D Drawings to ensure that BIM is being used to its fullest extent.
8.2	All 2D Drawings and Construction Documentation will adhere to and comply with the Singapore CAD Standard CP83. The Singapore CAD Standard CP83 is a set of guidelines and specifications for CAD practices in the construction industry in Singapore, including the standards for Drawing Format and Layout, Symbols and Notations, Layering and Line Weights Text and Dimensioning, etc.	CES BIM team will share Singapore CAD Standard CP83 and propagate TWRP drawing standards to all project teams and suppliers. Also, CES BIM team will do QA/QC on the drawing format for each drawing submission. BIM Manager will be responsible for the drawing's quality.

9	The Contractor shall be fully responsible for the items below	
9.1	For any scope or design change, the Contractor shall update the models.	CES BIM team will update model based on confirmed design input and record all model changes in the Model Change Log. In structural design, S.O. will issue an instruction upon Authority's approval and CES will update the models with the instruction.
9.2	For elements of the Works for which the S.O. completes design (e.g., structural), the Contractor is to maintain the BIM model and update the model with As-Built data	CES will be responsible for managing and updating the BIM model to reflect the actual construction progress and As-Built conditions.
9.3	For elements of the Works for which the Contractor is to complete the design (e.g., MEICA), the Contractor shall update the model with approved designs and provide an As-Built model along with the linked database upon completion	CES will perform quality checks and validations to ensure that the updated BIM model accurately represents the approved designs and reflects the As-Built conditions.
9.4	Any temporary works which include but are not limited to vehicle and crane movements, temporary facilities, temporary roads, temporary works, formworks, scaffoldings those activities which impact any other Contract or require permits.	CES will provide a model for temporary items to demonstrate that an activity is safe and/or does not impact other Contractors or within contracted work
9.5	Model of temporary structure as per supplier shop drawings, details, and method statements.	CES will update temporary structure into native model based on supplier information.
10	The Contractor shall comply with the requirements of BCA for statutory submissions to regulatory agencies, as mandated by the prevailing BCA BIM e-submission regulations.	CES will follow the BCA's BIM Essential guide for collaborative virtual design and construction. And CES will be responsible for liaison with BCA and other relevant regulatory agencies on all consultations and submissions related to the BIM development
11	P&ID Data	
11.1	The databases will be synchronized with P&ID drawings and shall be kept up to date with any Contractor changes. The P&ID	The database will be completed prior to the handover of the asset

	database shall also be kept up to date with any Contractor-designed elements.	
11.2	With suitably qualified persons to administer an SQL database and provide interrogation, issue resolution, and exporting of content from SQL.	Qualified person will be arranged to manage the SQL database. The P&ID database report will be submitted along with each version P&ID drawings for internal engineer team review and S.O. review.
11.3	Any new elements to add to the database shall seek approval from the S.O. P&ID information is a part of the LoI (held in databases) which will be provided by the S.O. upon contract award.	CES will submit Data Change Request to S.O. for approval if some new elements need to be added in the database (See Section 5.2.2).
12	Integration with Asset Management System (AMS)	
12.1	The Board has set up an AMS for the whole organization and, in addition to the BIM data, data shall be provided for AMS Integration	CES will provide computerized asset listing data/information to the Board's AMS for data integration before the end of the Contract
12.2	6D parameters shall be configured in BIM authoring software	As the construction progresses, CES will be responsible for collecting the As-Built data.
12.2.1	6D Data Drop 1: During asset/product detailed design or equipment requisition	CES will provide all details from final approved datasheet
12.2.2	6D Data Drop 2: During asset delivery, installation, or construction	CES will provide additional installation details, FAT test sheets, cost, 4D and 5D codes
12.2.3	6D Data Drop 3: During asset commissioning, start-up, and operation	CES will provide maintenance task details, SAT test sheets, O&M training & instructions, HAC register, confined space register
13	The deliverables and timing need to be reflected in the MIDP for the S.O. to review and verify BIM progress. The MIDP serves as a roadmap for information exchange and delivery throughout the project lifecycle.	CES BIM team will keep updating MIDP to clarify the model information exchange with other contracts and each BIM work result delivery schedule throughout the project lifecycle.
14	The specific requirements for the items listed for each data drop to be referenced	CES will ensure that the specific requirements for the items listed in each data drop are properly

	within the specifications in the tender documentation.	communicated to potential bidders, CES will reference these requirements within the specifications in the tender documentation.
15	Computerised asset listing data/information shall be provided for data integration to the Board's AMS before end of the Contract.	CES will validate the integrated data within the Board's AMS to ensure that it aligns with the contract requirements and meets the AMS's data standards.

6.1.2 Level of Assurance on BIM Requirements

The following table shows the level of assurance on BIM requirements. During BIM execution, CES will check if the BIM works meet the requirement in below three levels. Level 1, BIM Authoring Leads will check the integrity of BIM deliverables, to ensure the BIM files operate smoothly and the content is completed. Level 2, BIM Manager will check the standardization of the BIM results. Level 3, C4B project team, including BIM Manager, Engineering Manager, Quality Manager, etc, will evaluate the role of BIM and checked if its role can be maximized during the project implementation process.

Table 6-2 Level of assurance on BIM requirements

Assurance Level	Detailed	Person in charge
Level 1 Assurance the integrity of BIM deliverables.	Can the BIM files be opened properly? Is the content of the BIM files complete? Have all the elements in the BIM deliverables included complete data information?	BIM Authoring Leads
Level 2 Assurance of the compliance of BIM deliverables.	Are model element properties meet the requirement? Does the BIM work meet the BIM standard? Does the management follow the BEP, roles and responsibilities, and folder structure? Is the data accurate? Are the clashes solved?	BIM Manager
Level 3 Assurance BIM deliverables are helpful for the project construction	BIM Model walkthrough. Through the virtual to check is it buildable, design intent functional. Check the operation and maintenance access. Check the safety in design and during construction.	BIM Manager Engineering Manager Quality Manager

6.2 Quality Management

6.2.1 Overall Strategy

Quality Control (QC) check must be conducted by the responsible BIM Authoring lead. The positive QC result shall then be reported to the BIM Manager for record-keeping and process validation purposes.

QC checks should be carried out concurrently with the modelling process. A QC checklist will be prepared and used to ensure the BIM model's quality. The QC checks should be conducted by a staff member who is not the modeller of the BIM model. Additional QC checks may be conducted as deemed necessary by the BIM Manager.

All QC results should be properly documented by the QC checker. This documentation is essential to maintain the quality and accuracy of the BIM model and should be accessible to all relevant parties.

6.2.2 Quality Assurance/Control Procedure

QA/QC check shall be performed every time before BIM deliverables submission to ensure the quality of BIM deliverables. The modelling team shall prepare the models before submission. Then the BIM Authoring leads will check the quality. At last, after being reviewed by BIM manager, Engineering Manager, and Quality Manager, DC can forward the submission.

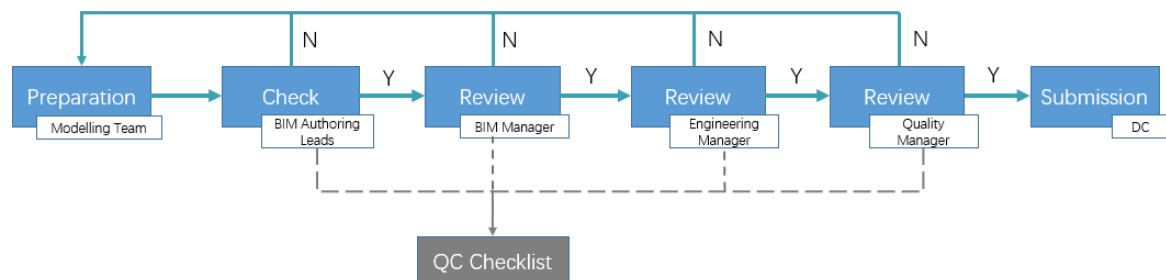


Figure 6-1 Quality Assurance/Control Procedure

6.2.3 Quality Assurance/Control Details

1) Quality control checks

Quality control checks are divided into native model QC checks, federated model QC check, drawing QC check, 4D QC check, 5D QC check, and P&IDs QC check. The check items contain most of the important information and data checking items that need to be accurate, which can refer to the checklists Appendix A.1~A.6, 131242-WST00-FM-IM-0001~0006 in the Appendix.

To comply with QA/QC checklists, CES will thoroughly review the QA/QC checklist to understand the specific requirements, standards, and processes outlined. This includes identifying the key checkpoints, activities, and documentation needed for compliance. BIM Manager should be the responsible individual in the team to oversee and implement the QA/QC checklist. CES will specify the BIM QA/QC check list and document the quality control procedure based on the master information delivery items in MIDP. Then the monthly VCM submission, milestone submission, and item in MIDP will strictly follow the checklist requirement. The QA/QC plan will follow BIM works delivery plan. Also, CES will ensure that all relevant personnel are trained on the QA/QC checklist requirements and the associated quality control procedures.

Before submission, the 3D models, 4D, 5D, and P&IDs should be first reviewed by BIM Authoring Lead, BIM Authoring Lead should check the BIM results item by item in the QA/QC lists. Modify the non-conforming items until all items meet the requirements. Then BIM Manager should check these items again to ensure there is no issue in the BIM format before sending to Engineering Manager/PCM/QS, and Quality Manager to check on the engineering item. The QA/QC Checklists should be attached to each BIM result before official submission. Also, to improve QA/QC, contractor can add some customized items to improve the quality. The following table shows the BIM quality control check details.

Table 6-3 BIM Quality Control Check

Checks	Definition	Responsible Party	Software Program(s)	Frequency
Visual check	Ensure there are no unintended model components, and the design intent has been followed	BIM Coordinator	Navigator Open Building Designer OpenPlant Modeller	As Required
Interference check	Detect problems in the model where two building components are clashing including soft and hard	Interface Manager, BIM Manager	Open Building Designer OpenPlant Modeller	As Required
Standards Check	Ensure that the BIM and CAD Standards have been followed (fonts, dimensions, line styles, levels/layers, etc.)	BIM Manager, Engineering Manager Quality Manager	Open Building Designer OpenPlant Modeller	As Required
Model integrity checks	Ensure that the data requirements are fulfilled, that there aren't undefined or duplicate elements	BIM Coordinator	Open Building Designer OpenPlant Modeller	As Required

- 2) According to the detail design, additional design, and design change, there will be many model changes. The changes will be recorded in the change log with the information including each model number, model version, detailed change contents, etc for tracking the changes. The change log will be reviewed in CES project team regularly to ensure all changes confirmed and meet the design and BIM requirements. The details of model change log register can be referred to Appendix A.10-131242-WSW41-FM-IM-0010.

- 3) Clash detection is required to identify all modelling and design clashes. With the proposed solution agreed with S.O., we should solve the clash and generate the “zero clashes” report, to ensure the model quality. The tolerances for clash detection should be set as 0 mm.

The clash detection should include the groups below:

- a) Architectural versus Structural;
- b) (Architectural + Structural) versus Civil;
- c) (Architectural + Structural) versus (Building Service);
- d) (Architectural + Structural) versus (MEICA); and
- e) Building Services versus MEICA.

The model clash detection summary is shown below and the quantities of clashes in each group need to be recorded.

Table 6-4 Clash Detection with C4B Summary

Facility	Building Level	Clash Group	Disciplines	Status

The clashes should be classified as major clash or minor clash following the definition as shown in the table 6-4 below. For minor clashes, CES will resolve internally. For major clasher, they should be reviewed by contractor designed team first and contractor to propose the resolution. The final resolution to the clashes needs approval from S.O.

Table 6-5 Classification of Clashes

Clash Type	Description
Major Clash	• Any clash causing time & Cost Impact
	• Interface with other contractors
	• Any case impacting structure
	• Any case impacting on Authority approval (such as impacting to ST approval)
	• Related Safety regulation and code compliance
Minor Clash	• Any clash not causing time & cost impact
	• A case can be done as field change
	• Rerouting small bore pipe at field (such as less than 100mm)

The clash check process can be referred to as Section 3.1. The clash report can be referred to 131242-WSW41-FM-IM-0007~0008.

7 Interface Management

7.1 Interface Management

For interface management, C4B will have an interface with C1A, C2A, C3A, C3D1, C3D2, C3B2, C4A and C5A contractors. The information exchange among different contractors will be based on the monthly VCM. CES BIM team will prepare VCM files including the latest models to submit for S.O. review. Then S.O. will confirm every month's submission and share the latest model to other contractors who need C4B project model information. Also, the CES BIM team will officially request regular share of other contractors' latest models from S.O. if needed.

Table 7-1 Interface Type With Other Contract

SN	CONTRACT	INTERFACE TYPE	FILE TYPE
01	IWMF	Piping	EX
02	C4A	Piping, water Drain, Panel , Cable Tray	i.dgn
03	C1A	Piping, Piping, JDUCT	i.dgn
04	C5A	Panel, Cable Tray	i.dgn
05	C3B1	Piping	i.dgn
06	C3A	EDUCT	i.dgn

BIM Manager is responsible for overseeing the implementation and coordination of BIM processes and technologies throughout the project lifecycle. BIM Manager interacts with external stakeholders, like S.O., and vendors, to coordinate information exchanges and align BIM processes with their requirements.

Engineers/specialists play a critical role in the interface coordination: (a) Help understand project requirements and review BIM models, drawings, and relevant documentation to gain a comprehensive understanding of the interface between contracts. (b) Identify interface points: Engineers/specialists need to identify the interface points where their discipline interfaces with other disciplines or systems within the BIM model. This involves analysing the model, identifying potential clashes or conflicts, and determining the critical coordination areas. (c) Collaborate with other disciplines: Engineers/specialists will collaborate with engineers/specialists from other disciplines to exchange information, share BIM models, and resolve clashes or conflicts. They should engage in regular coordination meetings to discuss interface issues, exchange knowledge, and find feasible solutions. (d) Propose design modifications: If design modifications are required to resolve interface issues, engineers/specialists should propose appropriate changes within the BIM model. They need to collaborate with the design team and relevant stakeholders to develop feasible solutions that align with project requirements. (e) Provide technical guidance: Engineers/specialists should provide technical guidance and expertise related to their discipline within the BIM environment. They should

ensure that interface coordination aligns with industry standards, codes, and best practices. This may involve reviewing design calculations, technical specifications, and construction methodologies within the BIM model. (f) Review and validate subcontractor models: Engineers/specialists should review and validate the BIM models produced by subconsultants, or subcontractors related to their discipline. This includes verifying that the models comply with project requirements, are accurately coordinated, and align with the overall project design.

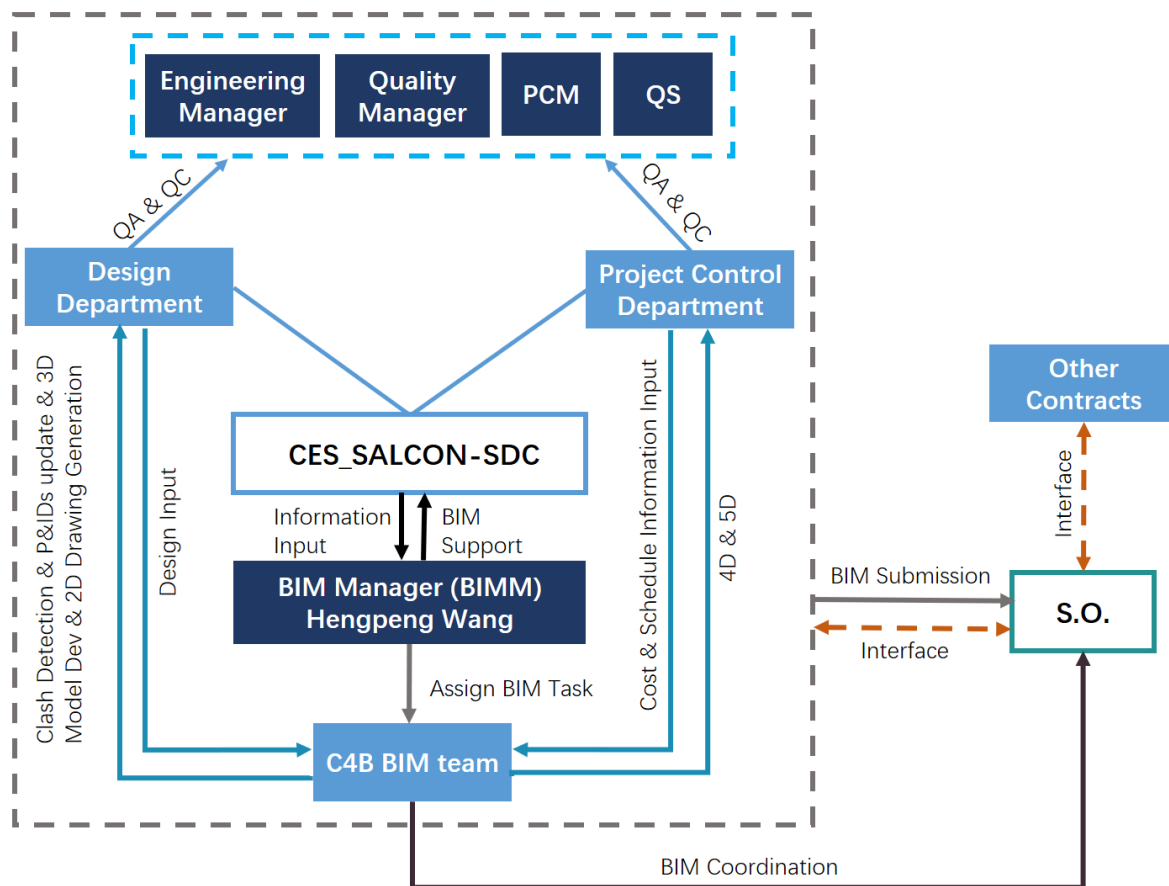


Figure 7-1 Interface Management Overview Map

7.2 Model Development Interface Management Workflow

During C4B BIM work execution, C4B BIM team will further identify the interfacing contracts and request from S.O. for monthly model sharing of related contracts. At the same time, C4B BIM team will update the latest design and input the latest design and construction information into the model for sharing with other contracts to reference. All TWRP contracts can realize the latest information sharing based on monthly model exchange. Then the model development interface management work will be carried out based on the monthly shared model.

As shown in Figure 7-2, CES Design Team will provide design input for model development. Then the updated model will be submitted to S.O. for review in monthly VCM. S.O. will receive the latest model from all TWRP contracts and share other contract models to those who need every month. Then CES BIM team will combine the model with received other package model to do clash detection

and have regular BIM walkthrough with CES engineer team together to check the issues with other contractor. The clashes and design issues will be raised in S.O. organized coordination meeting among different contracts. The resolution will be recorded officially and feedback to all parties in the meeting. Then the related model will be updated in next VCM based on the resolution.

During the procedure, OpenBuilding Designer will be used to review Architectural, Structural, Electrical, ICA design and OpenPlant Modeler will be used to review Mechanical design and model information. Also different contracts combined model review can choose Navigator as a tool and record the review comments. The clash detection can be also done by OpenBuilding Designer.

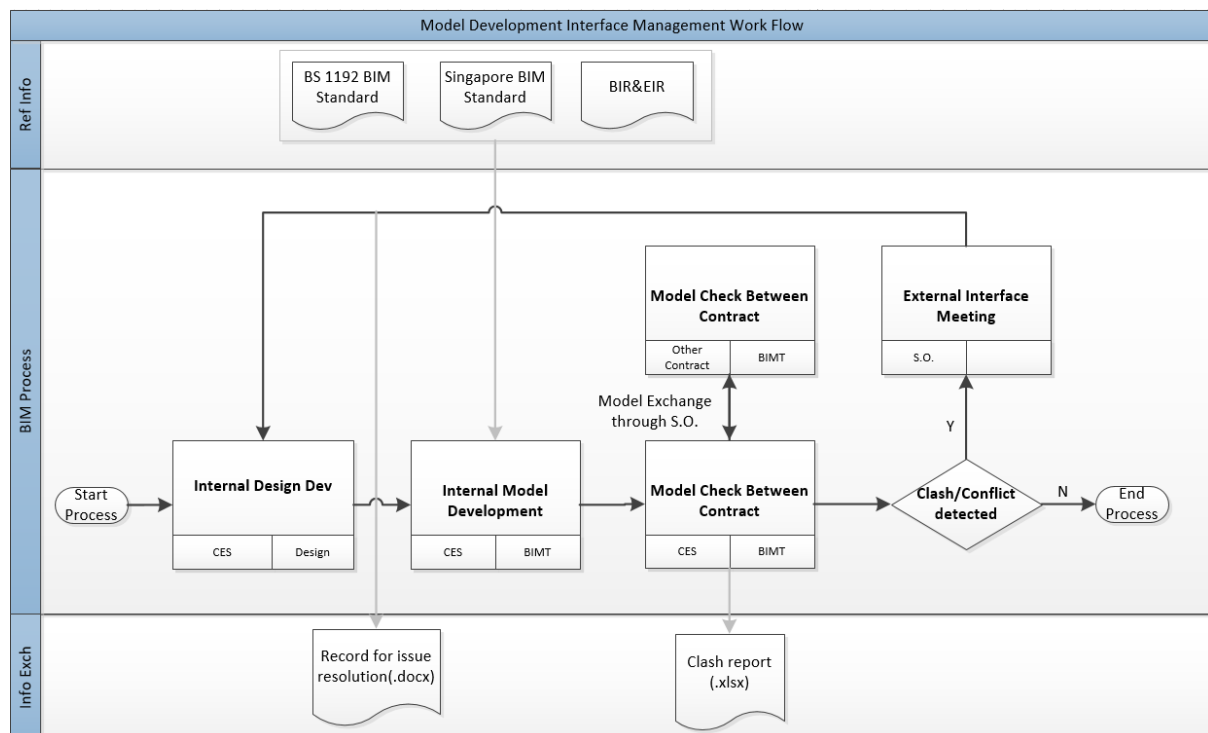


Figure 7-2 BIM Interface Management Workflow

Notes: In the square flow box, the top content is the related work, the bottom left corner content is the responsible company, and the bottom right corner content is the responsible person.

7.3 File Exchange with Interface contract

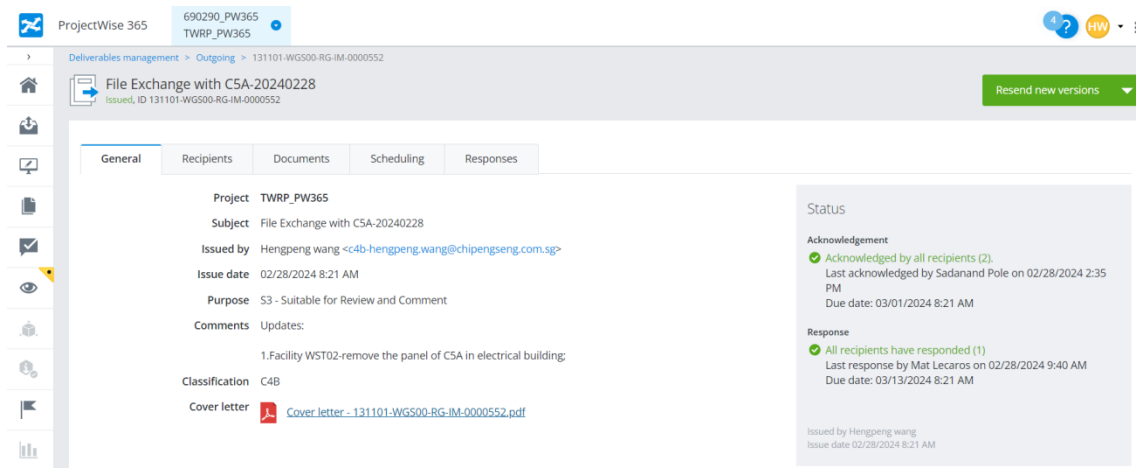
For BIM interface management, CES will have an interface with C1A, C3A, C3B1, C4A and C5A contractors. The information exchange among different contractors will be based on the monthly VCM. CES BIM team will prepare VCM files including the latest models to submit for S.O. review. Then S.O. will confirm every month's submission and share the latest model with other contractors who need C4B project model information. Also, CES BIM team will officially request regular share of other contractors' latest models from S.O if needed.

When it is necessary to organize the combined VCM walkthrough meetings with other contract, CES will share the latest federated i.dgn model to other contracts through ProjectWise 365. CES will also receive the latest federated i.dgn model of other contracts through ProjectWise 365.

Table 7-2 Interface Checking Items

ITEMS	DESCRIPTION
Version	CES BIM team will check the version and date to ensure received file are the latest
Change log and Opening registers	CES BIM team will check the change log and Opening registers to identify the impact of changes on C4B
Tie-in Points	CES BIM team will check the Tie-in Point tags in the model to ensure the connection for same Tie-in Points in different contract is correct
Clash Detection	CES BIM team will combine model with the interfacing contracts model to check the clash following the procedure in Section 1.1 Clash Detection and Resolution Workflow and then discuss the clash in coordination meeting
Combined Design Review	Based on the combined model, CES BIM team and engineer team will join coordination meeting with interfacing contracts engineer team and BIM team, to find and resolve the design issues

All the interfaces with other contracts should be managed by S.O., as well as the interface meeting. CES BIM team will provide any necessary documents to S.O. and attend the interface coordination meeting to support the interface work.

Figure 7-3 File Exchange with PW 365


7.4 Supply Chain

To ensure that suppliers are fully engaging with the contract, the contractor can use BIM in the following ways:

Collaborative design: CES BIM team will use BIM to help the design team collaborate with suppliers in the design process. Suppliers' actual model will be incorporated into the native model and reviewed by CES engineer team ensuring that suppliers' input is fully considered in the design. This will help to ensure that the suppliers are engaged with the project from the beginning.

Sharing BIM data: CES will share BIM data with suppliers to ensure that they have access to accurate and up-to-date project information. CES BIM team will give suppliers access to project CDE to get the latest model in time. The data from suppliers will be uploaded on ProjectWise by suppliers and reviewed by engineering team first. After confirmation BIM team will incorporate the suppliers' data into BIM model. Similarly, suppliers-related BIM model will be reviewed by engineering team and BIM manager first and then transferred to suppliers through ProjectWise. This will help to ensure that the suppliers are fully informed about the project and can make informed decisions.

Providing BIM training: CES BIM team will provide BIM training to suppliers to ensure that they have the necessary skills and knowledge to use BIM effectively, including ProjectWise, Open Building Designer, and Open Plant Modeler, to ensure they can smoothly get the latest project information from ProjectWise and use BIM authoring tools to carry out their work. Also, Technical meetings regarding BIM will be conducted between BIM/Engineering team and suppliers when needed. This will help to ensure that suppliers are able to engage with the project using BIM tools.

Encouraging BIM adoption: CES will require suppliers to adopt BIM tools and workflows by including BIM requirements in the contract documents. CES will also be specified EIR to suppliers to ensure their input totally follow the standard requirement. This will help to ensure that suppliers are fully engaged with the project and are using BIM to its full potential. For example, after the equipment design is approved, the suppliers will provide the equipment models which can be inserted into Bentley software and relevant equipment information (.xlsx/pdf) to input to BIM. The requirement of model LOD will be communicated with suppliers for them to prepare the models.

Managing the BIM Results from Supply Chain: Define clear protocols and guidelines for model management and clarify to supply chain, including outlining the file formats, naming conventions, and version control procedures to ensure consistency and easy identification of models; Align modelling standards and requirements with sub-consultants, subcontractors, and supply chain partners, to ensure that their models adhere to project-specific standards, enabling seamless integration and coordination; Establish effective communication channels to inform sub-consultants, subcontractors, and supply chain partners about any model revisions or updates, which include sharing BEP and MIDP, issuing formal notifications, conducting coordination meetings and so on; Give timely feedback to subconsultants, subcontractors, and supply chain partners regarding their models. Document the feedback and related communications to track progress, issue resolutions, and ensure accountability.

By using BIM in these ways, the contractor can ensure that their suppliers are fully engaged with the project and are using BIM to its full potential. This will help to ensure that the project is delivered on time, within budget, and to the required quality standards.

8 Technological Infrastructure

8.1 Software

The following table shows software, software version, and software function as per requirements in the 131101-WSW41-SP-SG-0102. Throughout the project, it is understood that the software version will change. So, if S.O. evaluates the newer version and determines the software version can be upgraded to take advantage of added features, and functionality that might increase the efficiency of the project, we will evaluate and discuss with S.O. to confirm.

Table 8-1 List of Software

Design/Modelling Software	Version	Function
Primavera	P6	Construction Scheduling
Microsoft Office	2016	All documentation
SQL Server Standard	2019	Hosting P&ID database
Bentley Open Buildings Designer	10.08. XX	Architectural, Structural + Design review/Clash Detection + 5D QTO + 6D COBie/AMS
Bentley Open Buildings Designer	10.08. XX	MVAC, Electrical, Plumbing, Fire Protection + Design review/Clash Detection + 5D QTO + 6D COBie/AMS
Bentley OpenPlant Modeller	10.09. XX	Process Mechanical, Instrumentation & Controls + 6D AMS
Bentley Open Plant P&ID	V8i SS10	P&ID - Process Mechanical, Instrumentation & Controls + 6D AMS
Bentley OpenRoads Designer	10.09. XX	Civil - Site Grading, Roads, Tunnel, Utilities, Geotech, and Site Analysis
Bentley View	10.15. XX	2D /3D Review
Bentley SYNCHRO 4D Pro	06.03.02	4D Sequencing

Adobe Acrobat	11.0+	2D Design Collaboration/Redlining
Bentley Open Buildings Designer	10.08. XX	3D design collaboration
Autodesk Navisworks	2019	Delivery of model to the Board/PM

8.2 Hardware

To fully utilized the BIM implementation of the project, below recommended hardware specification requirements.

Table 8-2 Hardware Specifications

Hardware	BIM Workstation
Desktop computer	
CPU	Intel® Core™ i9-9900K / i7 9700k
Memory	32GB RAM DDR4 2666+ or more
Display Card	NVIDIA Quadro P4000 or faster
Hard Drive	500+GB SSD & Min. 2TB 7200+RPM storage or lager
Network	Gigabit Ethernet Adapter with internet connection
Operating System	Microsoft® Windows® 10 Pro for Workstations 64 bits or later
Input Devices	16X DVD+/-RW with Dual Layer Write Capabilities or faster USB 3.0, 4 Ports with 2 ports in front or more Standard USB Keyboard with 104/105-key Optical USB Mouse with 2 buttons & Scroll Wheel
Output Devices	27-inch QHD Monitor x 2 or larger
Laptop	
Notebook	Dell Alienware m15 or faster Intel® Core™ i7-8750H

	NVIDIA® GeForce RTX™ 2080 8GB GDDR6 500+GB SSD 32GB RAM
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8.3 Other Infrastructure

At the beginning of the project, Bentley will carry out a separate training session for BIM team. The training is focused on the software or function.

During the project, BIM manager will carry out training for the modelers to strengthen the BIM skills. Also, BIM manager will organize meeting regularly for work experience exchange.

BIM team will carry out training for contractor and other consultants, the training is focused on the basic usage of the BIM tools and the understanding of the BIM models.

Table 8-3 Schedule of Training

S/N	TASK	LECTURER	PARTICIPANT	Date
1	ProjectWise Explore + Bentley Viewer Training 1st batch	BIM TEAM	Engineer Team, Subcontractor, Suppliers	01-Jun-23
2	ProjectWise Explore + Bentley Viewer Training 2nd batch	BIM TEAM	Engineer Team, Subcontractor, Suppliers	09-Jun-23
3	ProjectWise Explore Workshop-Resolve issue where subcontractors cannot upload	BIM TEAM	Engineer Team, Subcontractor, Suppliers	07-Jul-23
4	Bentley Workshop sessions 1	BENTLEY	BIM Team, Engineer Team	19-Jun-23
5	Bentley Workshop sessions 2	BENTLEY	BIM Team, Engineer Team	30-Jun-23
6	Bentley Workshop sessions 3	BENTLEY	BIM Team, Engineer Team	07-Jul-23
7	OBD	BENTLEY	BIM Team, Engineer Team, Subcontractor, Suppliers	07-Oct-23
8	Internal training of OBD、OPM(reference & review model)	BIM TEAM	BIM Team, Engineer Team	31-Aug-23
9	OPM sessions 1 (M & I)	BIM TEAM	BIM Team, Engineer Team, Subcontractor, Suppliers	01-Sep-23
10	OBD sessions 2 (E)	BIM TEAM	BIM Team, Engineer Team, Subcontractor, Suppliers	TBC
11	ProjectWise Explore+Bentley Viewer Training batches	BIM TEAM	BIM Team, Engineer Team, Subcontractor, Suppliers	TBC

9 Project Deliverables

It can be referred to Master Information Delivery Plan (MIDP) (Appendix A.9-131242-WSW41-FF-IM-0009) for the details of the 3D model delivery plan. This MIDP shows each 3D model file that needs to be delivered, the brief information of each model, and the planned issue date in every stage. CES BIM Team will update the model information and model updated schedule based on the design, construction, As-Built submission schedule continuously during the project. CES will transfer the latest MIDP to each party internally for all to track the progress and changes of the model in time. Also, CES BIM Team will submit MIDP to S.O. in the monthly VCM for review.

9.1 Deliverables during Design and Engineering

CES will develop design models following the EIR, corresponding also to the final approved amendment plans submitted to the relevant authorities.

Table 9-1 Deliverable During Design

No.	Deliverable	Formats
1	3D	
1.1	The updated native model (including all approved designs, all construction works completed or in progress, all approved changes)	.dgn
1.2	The updated Discipline Container Model (including all approved designs, all construction works completed or in progress, all approved changes)	.dgn, .i.dgn
1.3	The updated Federated & Contract Models (including all approved designs, all construction works completed or in progress, all approved changes)	.i.dgn , .i.bim ,
1.4	The updated interface contract model (including all approved designs, all construction works completed or in progress, all approved changes)	.NWD
2	4D	
2.1	The model(s) with embedded Programme for the Works. The model shall include temporary works, planned permitting works, interfaces with any other contractors as a minimum.	.sp for Bentley SYNCHRO 4D Pro Document, .xer/.xlsx for Primavera P6
3	5D	
3.1	The model(s) with associated exports of data for project expenditure tracking and for verification of monthly progress claims.	.i.dgn/.dgn for 5D Code embedded native model, .xlsx for BoQ Report
4	6D	

4.1	Updated models with As-Built information embedded for each section of work completed that month.	.ifc2x3 for BIM model with COBie/AMS; .xlsx for COBie Workbook
5	P&ID	
5.1	P&ID drawing sheets	.dgn , .i.dgn , .pdf
5.2	Datasheet report	.xlsx
6	Clash resolution including clash detection reports.	.xlsx
7	Laser scanning point cloud file, and visual verification report.	PTS; E57 or POD
8	A completed preparation and validation checklist for each model that the The contractor shall develop and provide.	.xlsx
9	Supporting CAD (2D) information in (.i.dgn) format.	.dgn, .pdf
10	Supporting or other required documentation in PDF format.	.pdf

9.2 Deliverables during Construction

The models to be submitted during construction as shown in Table 9-2 shall be updated monthly during the implementation of the project. The below items should be submitted to S.O for approval every 8th of the month except the 2D drawings.

Table 9-2 Deliverables during Construction

No.	Deliverable	Formats
1	3D	
1.1	The updated native model (including all approved designs, all approved changes)	.dgn
1.2	The updated Discipline Container Model (including all approved designs, all approved changes)	.dgn, .i.dgn
1.3	The updated Federated & Contract Models (including all approved designs, all approved changes)	.i.dgn , .i.bim ,
1.4	The updated interface contract model (including all approved designs, all approved changes)	.NWD
2	4D	

2.1	The model(s) with embedded Programme for the Works. The model shall include temporary works, planned permitting works, and interfaces with any other contractors as a minimum.	.sp for Bentley SYNCHRO 4D Pro Document, .xer/.xlsx for Primavera P6
3	5D	
3.1	The model(s) with associated exports of data for project expenditure tracking and for verification of monthly progress claims.	.i.dgn/.dgn for 5D Code embedded native model, .xlsx for BoQ Report
4	Clash resolution including clash detection reports.	.xlsx
5	A completed preparation and validation checklist for each model that the Contractor shall develop and provide.	.xlsx
6	Supporting CAD (2D) information in (. i.dgn) format.	.dgn, .pdf
7	Supporting or other required documentation in PDF format.	.pdf

9.3 Deliverables after Construction (As – Built)

CES will develop As-Built models following the EIR, that correspond to the final approved amendment plans submitted to the relevant authorities.

Table 9-3 Deliverables after Construction

No.	Deliverable	Formats
1	2D	
1.1	1 set of A1 size paper prints (drawings for statutory submissions shall be endorsed by the appropriate Professional Engineer, Singapore).	NA
1.2	1 soft copy of the As-Built Drawings in native “. dgn” format	.dgn
2	3D	
2.1	1 soft copy of As-Built BIM Models (Native File Format) updated with S.O approved contractor’s final design and embedded with COBie data to all elements identified by the Board as Facility Management (FM) and Operation & Maintenance items	.i.dgn
2.2	1 soft copy of the As-Built Model in “. i.dgn” format.	.i.dgn

2.3	1 soft copy of the As-Built Model in “. i.bim” format.	.i.bim
2.4	1 soft copy of the As-Built Model in Navisworks “.NWD” format.	.NWD
3	6D	
3.1	COBie exported Spreadsheets in excel format	.xlsx
3.2	AMS exported Spreadsheets in excel format	.xlsx
4	Laser scanning point cloud file, and visual verification report	PTS; E57 or POD
5	Updated database of P&ID data from OPPID	.xls, .mdb
6	Customized datasets and project configuration setup for BIM Authoring tools	NA

As-Built BIM Models and Drawings shall be endorsed “As-Built” and one set of original plotted drawings and submitted together with their soft copy in DVD-R media. All As-Built Models and Drawings in both hard and soft copies shall be submitted before Substantial Completion of the Works.

As-built BIM Models and Drawings will undergo a full review by the S.O. to ensure the As-Built data meets all requirements. Upon acceptance/approval, the As-Built BIM Models/Drawings and their storage media shall become the sole property of the Board. All related costs associated with complying with these requirements is deemed to have been allowed for in the tender, whether indicated explicitly or otherwise.

9.4 Asset Information Deliverable

CES will configure 6D parameters in the BIM authoring software and populate the parameters at the following suggested minimum data drop stages:

- 1) 6D Data Drop 1: During asset/product detailed design or equipment requisition;
- 2) 6D Data Drop 2: During asset delivery, installation, or construction; and
- 3) 6D Data Drop 3: During asset commissioning, start-up, and operation.

1 soft copy of As-Built BIM Models (Native File Format) updated with S.O approved contractor’s final design and embedded with COBie/AMS data to all elements identified by the Board as Facility Management (FM) and Operation & Maintenance items. The asset information deliverables are shown in below table.

Table 9-4 Asset Information Construction

No.	Deliverable	Formats
1.	Native BIM Models	
1.1.	All TWRP facility Models (“. dgn”)	.dgn

1.2.	Point cloud files	PTS; E57 or POD
1.3.	BIM-rendered scan files	NA
2.	2D	
2.1.	2D Drawing files (including native “.dgn” of all contracts drawings)	.dgn
2.2.	2D Adobe PDF	.pdf
3.	3D Federated Model Files	
3.1.	One Federated Model “.dgn”, “.i.dgn,” , and “.bim” of the respective contract scope with all linked information included	.dgn, .i.dgn , and .bim
3.2.	Facility level Federated models “.dgn”, “.i.dgn,” of the respective with all linked information included	
3.3.	Navisworks “.nwd” with all linked information included	.nwd
4.	4D	
4.1.	Bentley SYNCHRO 4D Pro : “.sp”	.sp
4.2.	Navisworks : “.nwd” and “.nwf”	.nwf
4.3.	Primavera P6 : “.xer” or “.xlsx”.	.xer or .xlsx
5.	5D	
5.1.	5D Code embedded in native model files	.dgn
5.2.	BoQ report from models for project expenditure : “.xlsx” format	.xlsx
6.	6D	
6.1.	BIM Model with COBie/AMS data in “.ifc2x3”	.ifc2x3
6.2.	COBie (.xlsx format) workbook	.xlsx
6.3.	Asset Management System(AMS) – as per the spreadsheet (“.xlsx” format)	.xlsx
7.	OPPID Database	NA